

Radiative cancellations and localized linear modes in three-dimensional sextic Q-balls

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Abstract

We study localized linear excitations of a complex scalar Q-ball in three spatial dimensions with potential $\mathcal{U}(S) = S - S^2 + S^3/2$. The linearized problem couples two frequency sidebands. In a region with one propagating and one evanescent exterior channel, radiation can cancel at particular parameter values. We assemble computer-assisted existence arguments for two radial modes and one dipole mode, together with a numerically observed sequence of fifteen radial candidate locations. The existence statements concern the infinite radial domain, using analytic exterior estimates and ball-arithmetic interior enclosures; they are not inferred from small finite-box linewidths. A thin-wall phase-matching description, partly calibrated, describes the observed near-uniform spacing in inverse detuning. Sequential and reportedly author-blinded comparisons test predictions for further linewidth minima, but do not certify the complete sequence or an infinite ladder. We distinguish these linear results from nonlinear confinement: a direct expansion produces second-harmonic sources in open channels, including monopole and quadrupole channels for a real dipole excitation. A numerical forced-response calculation supports nonzero outgoing amplitudes for an approximate dipole input. A complete monopole error budget additionally separates one second-harmonic coefficient from zero, conditional on the stated exact-mode and analytic enclosure hypotheses. This supplies an obstruction to a specified uniformly localized smooth continuation of a real dipole; circular polarization remains outside this monopole conclusion. The manuscript makes no claim of linear or nonlinear stability, a particle model, or an experimental realization.

Status. This is a complete working draft, not a submitted or journal-refereed article. The proof components have project-internal cross-house reviews; the assembled manuscript has its own narrower editorial checks. A second independent interval implementation is not provided. Numerical values are dimensionless throughout.

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1 Question and scope

A long-lived oscillation and an exactly localized linear mode are different objects. A numerically small resonance width may result from a cancellation, a reflecting boundary, or insufficient resolution. Conversely, even a localized linear mode need not extend to a localized finite-amplitude oscillation: nonlinear harmonics can couple to the continuum. Our objective is to separate these statements within a single specified scalar field model. We report three concrete existence enclosures, a broader numerical sequence, and an approximate mechanism for its spacing. The sequence index n is a continuation label, not a separately certified node count or a statement that no intermediate modes exist.

A limited mechanical analogy is a bell with several vibration patterns: exciting a particular pattern differs from exciting an arbitrary disturbance. Here the additional ingredient is cancellation of the outgoing amplitude in an available radiation channel, a general interference mechanism discussed in Ref. [4]. The cancellation can be lost under perturbations that do not preserve its condition. The analogy concerns linear modes; it implies neither undamped finite-amplitude ringing nor immunity to environmental disturbances.

The central contribution claimed here is this combination of model-specific linear existence enclosures and numerically tested spacing structure. We do not claim that embedded modes, interference suppression, Q-ball excitations, or thin-wall approximations are new concepts. No priority claim is made for the general mechanism. The nonlinear analysis is retained as a limit on interpretation, not used to advertise stable particles.

1.1 Relation to Q-ball excitations and continuum trapping

Q-balls belong to the established theory of localized scalar configurations carrying a conserved global charge [1]. Their perturbations and nonlinear interactions have an extensive history. Battye and Sutcliffe simulate collisions, charge transfer and fission in one, two and three spatial dimensions in the same classical sextic model, after unit conversion [13]. Their kinetic normalization is $\frac{1}{2}|\partial\phi|^2$ and their potential is $2\mathcal{U}$; $x = 2x_{\text{BS}}$, $t = 2t_{\text{BS}}$ give $\omega = \omega_{\text{BS}}/2$. Integrated energies and charges must also be converted in the chosen spatial dimension. Their slowly decaying one-dimensional breathing example is a dynamical precursor, not a certificate of an embedded linear mode. The model itself is not new here. Its perturbations require a coupled frequency problem. Kovtun, Nugaev, and Shkerin formulate the complex-field linearization with conjugate coupling and the two frequencies $\omega \pm \rho$ [2]. Their three-dimensional partial-wave equations provide a direct structural comparison, while the vibrational modes selected in their piecewise-parabolic example satisfy $\omega + \rho < 1$. That calculation does not establish the embedded modes of the sextic potential studied here.

Azatov, Ho, and Khalil study three-dimensional scattering in the same sextic model family [7]. With their $\phi_Q = \sqrt{2}f$ and $g = \beta = 1/2$, their linear coefficients U, W equal our D, C . Conjugating their carrier convention and reducing the radial functions maps their Appendix B partial-wave equations to Eqs. (4)–(5). The radial operator is established prior work; the question here is the existence of the particular localized solutions enclosed below.

The same work also treats Friedberg–Lee–Sirlin Q-balls containing a complex and a real scalar. Its Section 3 derives the coupled charged sidebands and neutral channel, their asymptotic thresholds, and scattering relations [7]. Thus an additional real field and a three-channel linearization are not new ingredients. These scattering calculations do not provide the particular infinite-domain embedded-mode enclosures examined here.

Zhang, Zhou, and Zhu obtain analytic scattering solutions for stepwise approximations to Q-balls of the same sextic family [8]. With their $g = \beta$ and $f_0^2 = 1/(2\beta)$, their interior wavenumber $\sqrt{-\rho_1}$, Eq. (101), equals k_{in} in Eq. (9). Their propagating scattering regime requires both exterior channels to be open. The common interior dispersion and thin-wall phase structure are therefore established ingredients; they do not themselves establish cancellation of the remaining open-channel amplitude in the one-open-channel regime considered here.

Ciurla, Dorey, Romańczukiewicz, and Shnir study a sextic Q-ball model in $1+1$ dimensions [3]. Their separation of bound, half-propagating, and quasinormal modes is particularly relevant: in the interval $1 - \omega < \rho < 1 + \omega$, one asymptotic channel propagates and the other is evanescent. They exhibit a small but nonzero radiation tail associated with a long-lived quasinormal mode and investigate nonlinear relaxation. This distinguishes a narrow resonance from the exact vanishing of the propagating tail required here. Their one-dimensional examples do not supply the spherical profile, origin conditions, or infinite-domain enclosure used in the present three-dimensional problem.

Evslin et al. analyze paired Floquet modes of small-amplitude Q-balls in $1+1$ dimensions [10]. Their counterrotating analysis relates a discrete component coupled to a propagating partner to a Feshbach-type quasinormal mode. The small-amplitude approximation and one-dimensional profile do not supply the full spherical sextic problem or the vanishing open-channel amplitude examined here.

Continuum trapping through channel interference is also an established mechanism. Friedrich and Wintgen relate interfering resonances to a vanishing decay width [4]. A rigorous related construction by Yu and Lu treats three Schrödinger channels on a half-line, with one open channel, a crossing of two uncoupled bound levels, and suitable nonzero radiation couplings [5]. Their small-coupling theorem concerns a potential that is exactly constant outside a finite interval. It is not directly applicable to the frequency-dependent two-channel Q-ball pencil, whose coefficients derive from a nonlinear profile and retain an infinite tail. We therefore use these works as comparisons, without assigning the specific Friedrich–Wintgen mechanism to our modes solely

from a vanishing radiation amplitude.

Resonant total reflection through coupled propagating and evanescent channels is established for discrete breathers [19], optical solitons [20], and spinor-condensate domain walls [21]. These are scattering solutions with an incident wave; localization of a closed-channel component is not by itself an embedded eigenmode of the full problem. Nor does the existence of open and closed channels suffice for reflection: the fully coherent homogeneous optical example remains transparent [20]. These comparisons motivate a broader channel-interference interpretation, without making an antiparticle branch a universal prerequisite or establishing a necessary condition on Krein signatures.

Accumulation of embedded solitons also predates this work. Malomed et al. derive a WKB/Bohr–Sommerfeld description in a quadratic optical model, with a predicted scaling $\varepsilon_n \sim n^{-6/5}$ [11]. Their matching is explicitly nonrigorous and its leading prefactor differs from their numerical fit. Their objects are nonlinear embedded solitons; ours are linear internal modes on changing Q-ball backgrounds. Neither the scaling parameters nor their exponents can be identified merely from the existence of a sequence of radiation cancellations.

Soffer and Weinstein prove nonlinear radiation damping under a positive Fermi-golden-rule hypothesis for a real cubic Klein–Gordon model [12]. Its initial discrete mode lies below the continuum and its third harmonic radiates. Their theorem does not directly cover our embedded mode, rotating background, or second-harmonic source. Our conditional continuation obstruction is not a dynamical decay theorem.

Suppressed radiation also occurs for real-field oscillons. Zhang et al. report multiple decay-rate minima for the ϕ^6 potential in Section 7.5 [9]. Their analysis concerns long-lived nonlinear configurations with radiative decay, rather than an enclosure of an exact embedded eigenmode. The authors also identify limitations in lifetime estimates near decay-rate minima. A small radiation signal or a long-lived oscillation is therefore not, by itself, the criterion used for the linear localization claims here.

An especially close comparison for the nonlinear radiation problem is provided by Inagaki and Murakami [23]. They numerically resolve five zeros of the second-harmonic source overlap for an internal mode of a thermal critical bubble, including three on the thinner-wall side. The fundamental mode is below the continuum; it is the leading second-harmonic radiation amplitude that vanishes. At a selected zero, third-harmonic radiation remains and controls the leading small-amplitude decay. These results neither constitute a linear embedded eigenmode nor establish a radiation-free nonlinear solution. They also do not establish an infinite sequence or its asymptotic spacing. Our linear localization claim concerns the fundamental mode itself; this distinction does not by itself establish priority.

The contribution examined here is the existence of particular localized solutions of that full linearized Q-ball problem at the stated parameter boxes, with regular origin conditions and controlled infinite tails. The literature above does not replace those checks, and this comparison does not establish priority. A finite collection of certified points also does not imply an infinite ladder, or that modes on distinct backgrounds belong to one fixed Q-ball spectrum. Finally, linear localization alone does not establish an exactly localized nonlinear oscillation of finite amplitude or nonlinear stability.

2 Field model and two-channel linearization

With metric signature $(+, -, -, -)$, define

$$\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - \mathcal{U}(|\phi|^2), \quad \mathcal{U}(S) = S - S^2 + \beta S^3, \quad \beta = \frac{1}{2}. \quad (1)$$

The free scalar mass is one. A spherical stationary background is $\phi_0 = e^{i\omega t} f(r)$ with real f , satisfying

$$f'' + \frac{2}{r} f' = (1 - \omega^2) f - 2f^3 + 3\beta f^5, \quad f'(0) = 0, \quad f(r) \rightarrow 0. \quad (2)$$

For definiteness the charge sign is chosen so that $Q = 2\omega \int f^2 d^3x > 0$ when $\omega > 0$. The thin-wall frequency threshold is $\omega_c^2 = \min_{S>0} \mathcal{U}(S)/S = 1 - 1/(4\beta) = 1/2$.

Use a real orthonormal spherical-harmonic basis and write

$$\delta\phi = e^{i\omega t}[a(r)e^{i\rho t} + b(r)e^{-i\rho t}]Y_{lm}(\hat{x}), \quad A = ra, \quad B = rb, \quad (3)$$

for real-frequency, real-radial-function modes. In a complex harmonic basis the second sideband instead carries Y_{lm}^* . Let $S = f^2$, $D = 1 - 4S + 9\beta S^2$, and $C = -2S + 6\beta S^2$. The radial equations are

$$A'' = \left[D + \frac{l(l+1)}{r^2} - (\omega + \rho)^2 \right] A + CB, \quad (4)$$

$$B'' = CA + \left[D + \frac{l(l+1)}{r^2} - (\omega - \rho)^2 \right] B. \quad (5)$$

Regular solutions satisfy $A, B = O(r^{l+1})$ at the origin. In the range $1 - \omega < \rho < 1 + \omega$, define

$$k^2 = (\omega + \rho)^2 - 1 > 0, \quad \kappa_c^2 = 1 - (\omega - \rho)^2 > 0, \quad \kappa_0^2 = 1 - \omega^2 > 0. \quad (6)$$

The A channel is open and the B channel closed. A localized mode requires cancellation of both oscillatory exterior components of A and exclusion of the growing component of B . Throughout, “embedded” means a localized solution in this explicitly open-channel regime. A separate operator-domain and essential-spectrum theorem for the full time-evolution generator is not asserted.

Spatial localization does not require compact support. A scalar component of laboratory frequency E with $|E| < 1$ has the vacuum decay scale $\kappa = \sqrt{1 - E^2}$ and free decaying asymptotic proportional to $e^{-\kappa r}/r$, with angular-momentum corrections. The threshold $|E| = 1$ requires separate treatment. For $|E| > 1$ a propagating channel exists, but its oscillatory exterior coefficients vanish for the localized linear modes considered here; exponentially forced tails can remain in the coupled problem. Infinite spatial support implies neither an unsuppressed long-range force nor a universal interaction. A stationary tail is also distinct from the causal propagation of a new disturbance.

2.1 Spatial mode shapes and phase-resolved reconstruction

The three-dimensional shape is obtained from the computed radial functions and the specified spherical harmonic, not from a point-particle picture. For real radial components and the positive-carrier convention of this paper, we display the rotating-frame perturbation per unit small amplitude,

$$\psi_1(\mathbf{x}, t) = Y_{l0}(\hat{\mathbf{x}}) \left[\frac{A(r)}{r} e^{i\rho t} + \frac{B(r)}{r} e^{-i\rho t} \right]. \quad (7)$$

The full first-order field is $e^{i\omega t}(f + \epsilon\psi_1)$. Thus a stationary background density does not mean that the complex field is static. A localized nonzero oscillation can have zero linear radiation; this is distinct from removing the oscillation itself.

Figures 1 and 2 use stored numerical reconstructions near C0 and T2. They are not pointwise interval-certified eigenfunctions and are not a nonlinear three-dimensional time evolution. The T2 arrays were produced in the conjugate carrier convention; the time factors in Eq. (7) conjugate that convention consistently. Interpolation is used only for display. All lengths are dimensionless model coordinates; no particle diameter or laboratory length is assigned.

A spatial node of ψ_1 is not the BIC cancellation condition. The latter is a condition on the frequency and background parameters. To visualize this distinction, Fig. 3 shows an archived C0 search using a normalized backward-Jost origin residual $F = (F_1, F_2)$. Its complex representation $F_1 + iF_2$ is a device for tracking the winding of two real boundary residuals, not the phase of the physical field. The finite-radius numerical diagnostic predates and does not replace the existence enclosure in Section 6.1.

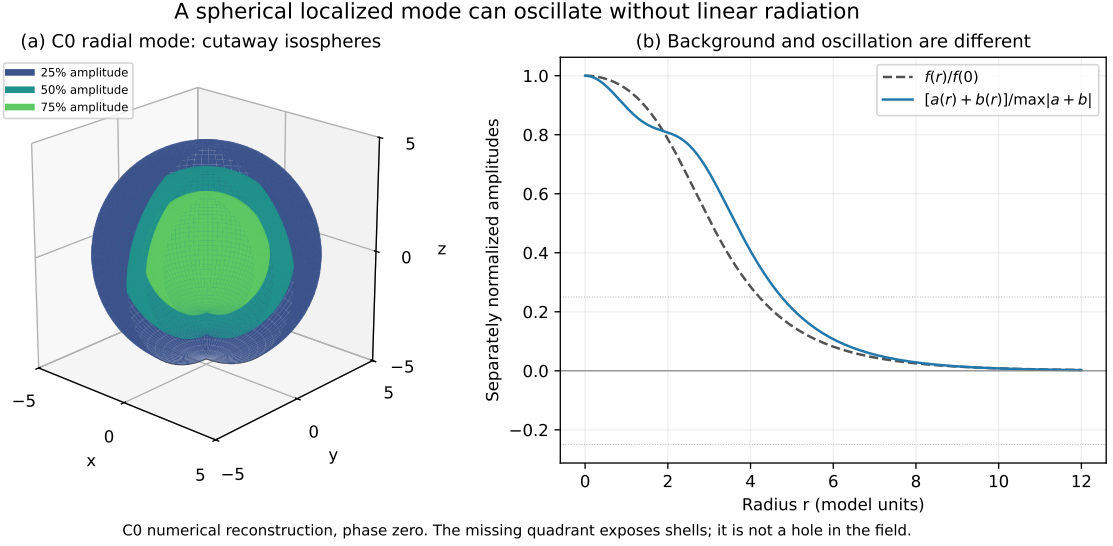


Figure 1: C0 spherical mode at phase zero. Left: cutaway surfaces of constant $|\psi_1|$ at 25%, 50% and 75% of its displayed maximum; color identifies the amplitude level. Removing a quadrant exposes the radial shells and is only a drawing choice. Right: the background and the radial mode, separately normalized for comparison. The shells are level sets, not material surfaces. These are numerical reconstructions associated with C0, not new existence certificates.

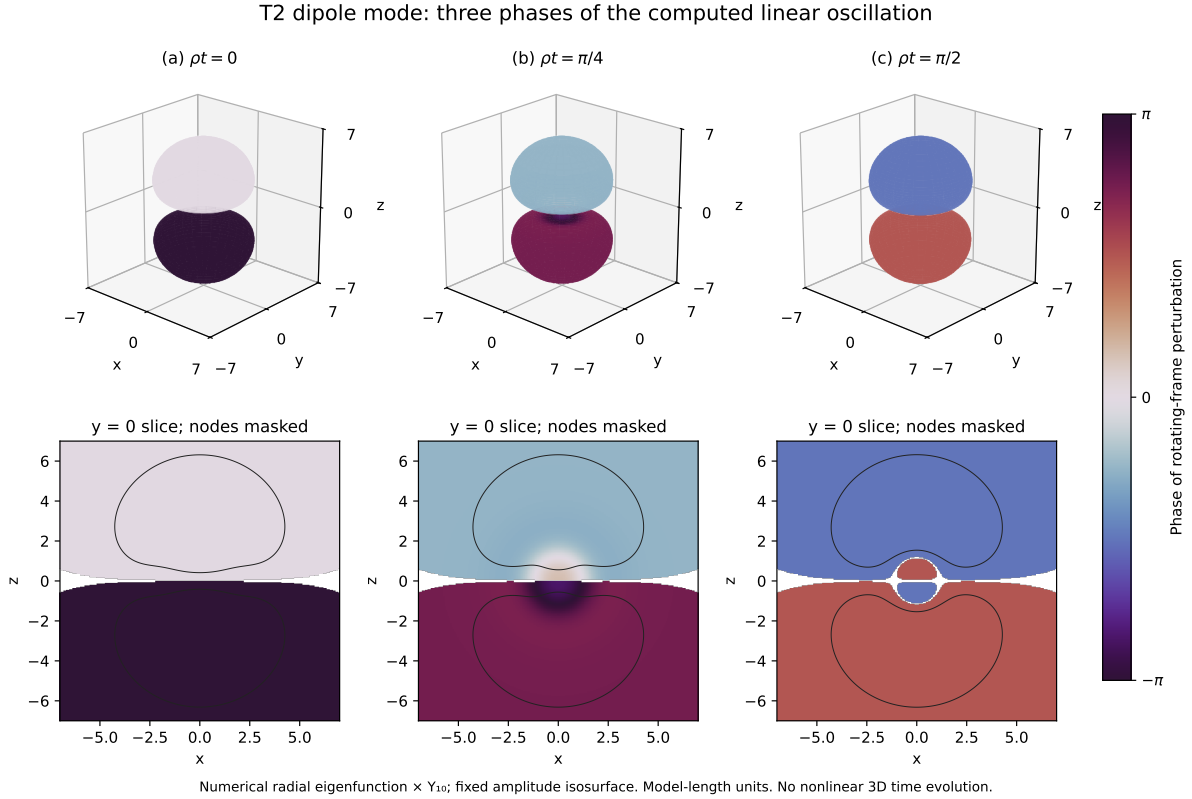


Figure 2: T2 with a real $Y_{10} = \sqrt{3/(4\pi)}z/r$ angular factor. Top: three phases of the linear mode at a common amplitude-isosurface level. Bottom: the corresponding $y = 0$ sections; phase is masked below 1.5% of the common displayed peak, because phase is undefined at a node. Color encodes $\arg \psi_1$ in the rotating frame, with a fixed common gauge and cyclic scale; it is neither a vector direction nor an additional charge. The opposite lobes are in antiphase. The changing quadratures follow the two computed radial sidebands and their analytic time factors, rather than a fresh nonlinear simulation.

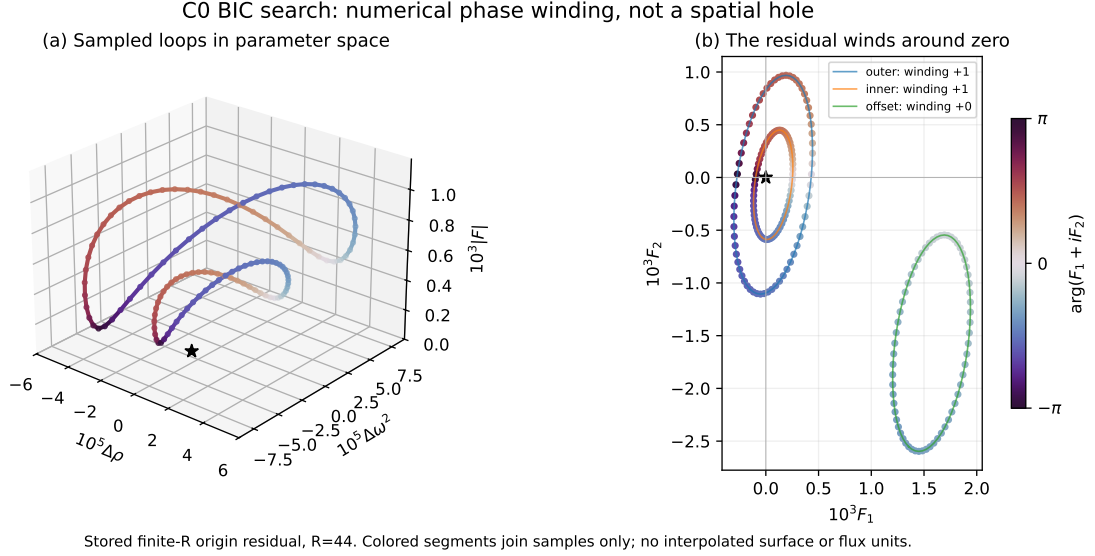


Figure 3: The C0 cancellation in parameter space. Left: two sampled parameter loops lifted by $|F|$, colored by $\arg(F_1 + iF_2)$; the star marks the numerical root. The reference coordinates are $\rho_{\text{ref}} = 1.744625038252254$ and $\omega_{\text{ref}}^2 = 0.7976953125$. Right: their residual images and an offset control, with numerical windings $+1, +1, 0$. Lines connect stored samples; no two-dimensional residual surface has been inferred from these loops. The historical calculation uses outer radius 44. The plotted residual is neither radiated power nor an absolute outgoing-wave amplitude, and the third plotting coordinate is not a third physical-space dimension.

3 Evidence hierarchy and numerical sequence

We separate three levels: a width minimum is a numerical candidate; a resolved winding test of a specified two-component matching map gives additional numerical evidence; an interval enclosure with analytic tails supports an infinite-domain existence statement. These labels cannot be interchanged. A complex pole search and a real matching zero are also not identical numerical tasks.

Table 1 distinguishes the continuation locations by their estimator and evidence class. Rows 1 and 2 use the subsequent existence certificates C0 and C1. Rows 6–10 now use the subsequent R13 calculation; the other numerical rows retain their archived data-package records. Rows 3 and 4 have resolved *numerical* winding tests in the truncated radial problem, not interval-certified winding enclosures. Row 5 has sign-change evidence, but its sampled winding test did not meet the resolution criterion. Historically, for indices 7–9, pole-width interpolation replaced a problematic direct sign calculation with strong interior growth. For indices 7–9 and 11–15, locations were reconstructed from short linewidth samples using signs selected to match neighboring slopes; those archived records did not report resolved numerical BIC winding tests. The old row-10 reference was an unrefined continuation estimate, distinct from the observed width depression at the scan edge.

R13 adds real matching sign changes and resolved numerical windings for $n = 6, 7, 8, 9, 10$ in the truncated radial problem. On both radial steps $h = 0.04$ and 0.02 , the windings for $n = 7, 8, 9, 10$ are respectively $-1, +1, -1, +1$, with maximum sampled phase increments below 0.301 radians. The $n = 1$ control agrees with the certified location to its prescribed tolerance; the $n = 6$ control gives approximately $\omega^2 = 0.569360$ and a resolved positive winding. This is new numerical evidence, not additional interval-certified existence results. For $n \geq 11$, the earlier linewidth evidence retains its previous status.

The R13 locations are linearly interpolated between ω^2 samples separated by 2.5×10^{-4} . Agreement between the two radial steps within 2.3×10^{-8} is therefore not a location error bound

of that size. The report’s interpolation-uncertainty estimate of a few 10^{-6} is not a validated enclosure. Signed-width reconstructions differ from these locations by at most 2.2×10^{-6} , but share profiles, equations and implementation with the matching calculation; this is not an independent replication. Candidate interior growth and window-maximum growth are different diagnostics, and the adopted growth threshold is a numerical convention, not an analytic error bound.

Table 1: Radial sequence at $\beta = 1/2$. Approximate candidate locations are rounded from the indicated archived records or new R13 calculation; displayed digits do not specify confidence intervals or validated location errors. For C0 and C1, the full enclosures are given in Section 6.

n	ω^2	Evidence and source record
1	0.797676787...	Existence enclosure C0
2	0.685128904...	Existence enclosure C1
3	~ 0.631450	Resolved numerical winding; A03
4	~ 0.601421	Resolved numerical winding; A04
5	~ 0.582417	Sign change / curve refinement; A05
6	~ 0.569360	Resolved numerical winding; R13-K2 ^a
7	~ 0.559848	Resolved numerical winding; R13-n7
8	~ 0.552619	Resolved numerical winding; R13-n8
9	~ 0.546940	Resolved numerical winding; R13-n9
10	~ 0.542364	Resolved numerical winding; R13-n10 ^b
11	~ 0.53860	Linewidth reconstruction; A11
12	~ 0.535449	Linewidth reconstruction; A12
13	~ 0.532772	Linewidth reconstruction; A13
14	~ 0.530470	Linewidth reconstruction; A14
15	~ 0.528469	Linewidth reconstruction; A15

^aHistorical A06 evidence, not the new R13 result: two refinement runs ended near 0.569397 and 0.569379, differing by about 1.8×10^{-5} . Neither reached the prescribed $|\text{Im } \rho| < 10^{-8}$ refinement criterion; convergence of the internal pole search does not establish convergence of the location refinement. These two estimates do not define a validated error interval.

^bHistorical A10 evidence, not the new R13 result: the scan showed a width depression at its edge $\omega^2 = 0.5425$; the former reference ~ 0.5424 was a continuation estimate, not a previously refined pole-width minimum.

4 Sequential predictions and their limits

Two prediction procedures were compared to the same target calculations. The first extrapolated approximately constant inverse-detuning steps and was updated after new results: first $n = 11$, then $n = 12, 13$, then $n = 14, 15$. It was not a single frozen five-point forecast. The second, a thin-wall model fitted using indices 5,7,8,9, was reportedly blinded to later results at the author level. Some target results were already known to the coordinating team. This is information separation, not a claim of global chronological preregistration or an independent numerical replication. The frozen thin-wall forecast also included $n = 10$ at 0.542382 ± 0.000020 . The new R13 interpolated location ~ 0.542364 is numerically compatible with this window. The old ~ 0.5424 reference was a continuation prediction, not a refined pole minimum. The new comparison supplies neither a validated location error nor a claim of exact zero linewidth. It also does not create retroactive global blinding: the R13 code author’s $n = 7$ prediction followed a positive smoke test, and is distinct from the coordinating team’s earlier forecast.

The sequential location rule used one stated window; the thin-wall comparison used its stated three-window rule. The handling of a failed linewidth-threshold check also changed in the last sequential stage, from failure to indeterminate, as documented before that stage. All five reported comparisons passed their respective rules. The thin-wall predictions lie systematically about 2×10^{-5} above the reconstructed locations. This is a visible model discrepancy, not five independent random errors. The first sequential prediction lies above its target, whereas the later ones lie slightly below. Neither successful comparison establishes a zero linewidth.

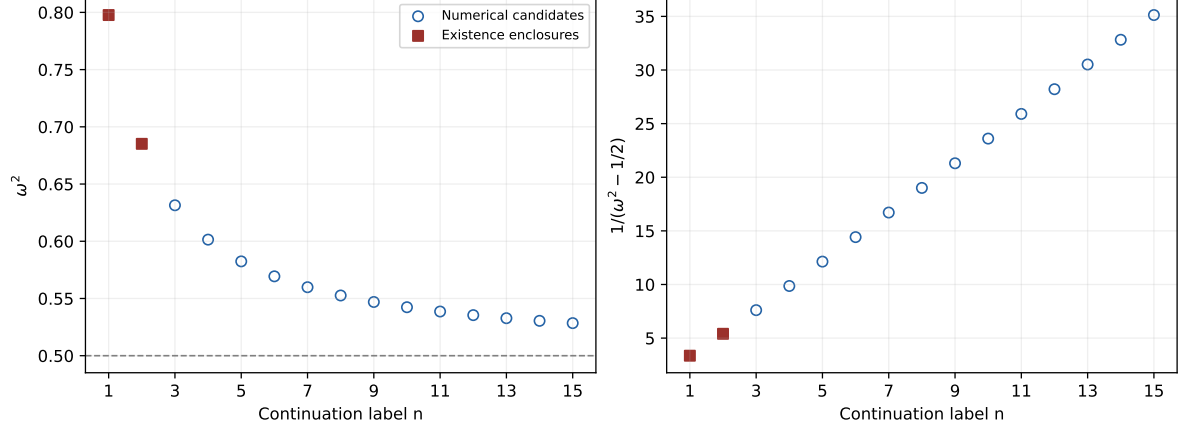


Figure 4: Reported radial locations and inverse detuning $1/(\omega^2 - 1/2)$. Filled symbols distinguish the two radial locations with existence enclosures from the remaining numerical candidates. No error bars are invented for rows lacking defensible intervals. The plot uses the new R13 interpolated locations for $n = 6-10$ ($\sim 0.569360, 0.559848, 0.552619, 0.546940, 0.542364$), archived estimator values for the other numerical rows, and C0/C1 from their certificates. R13 supplies numerical sign-change and winding evidence, not new existence enclosures. Inverse detuning does not improve location accuracy; this is neither a new fit nor a proof of accumulation.

Table 2: Reported locations and prediction windows. The $\pm$ values are preassigned diagnostic windows, not statistical confidence intervals. The two procedures used different acceptance rules.

n	Numerical location	Sequential extrapolation	Thin-wall prediction
11	~ 0.53860	0.53867 ± 0.00030	0.538619 ± 0.000025
12	0.535449	0.53544 ± 0.00012	0.535468 ± 0.000030
13	0.532772	0.53276 ± 0.00012	0.532792 ± 0.000035
14	0.530470	0.530460 ± 0.000040	0.530490 ± 0.000040
15	~ 0.528469	0.528458 ± 0.000050	0.528489 ± 0.000050

5 A thin-wall phase-matching description

Put $\epsilon = \omega^2 - \omega_c^2$ and $S_c = 1/(2\beta)$. The leading thin-wall profile and radius in three dimensions are

$$R_{\text{tw}} \simeq \frac{1}{2\sqrt{\beta}\epsilon}, \quad f^2(r) \simeq \frac{S_c}{1 + \exp[(r - R_{\text{tw}})/\sqrt{\beta}]}. \quad (8)$$

These are asymptotic approximations, not substitutes for the certified profiles. The leading radius relation is established prior work: with $\Delta = m_\phi^2 - \omega_0^2$, Heeck et al. use $\kappa^2 = (\omega^2 - \omega_0^2)/\Delta$ and $R^* = \sqrt{\Delta}R$, and obtain $R^* \simeq 1/\kappa^2$ in their Eq. (40) [22]. Here $\Delta = 1/(4\beta)$, which gives the radius in Eq. (8). Their inflection-point radius and our half-density radius need not have identical subleading coefficients. Diagonalizing the constant interior channel matrix gives the propagating interior wavenumber

$$k_{\text{in}}^2 = \omega^2 + \rho^2 - D_c + \sqrt{4\omega^2\rho^2 + C_c^2}, \quad D_c = 1 + \frac{1}{4\beta}, \quad C_c = \frac{1}{2\beta}. \quad (9)$$

At the stated interior amplitude, this is the same dispersion branch as Eq. (101) of Zhang, Zhou, and Zhu [8]; the interior wavenumber is an established ingredient, not a new result here. A regular interior standing wave reaches the wall with phase $\Psi = k_{\text{in}}R_{\text{tw}}$. The bare closed-channel wall approximation supplies a localized state, while matching it to the open interior wave permits a sign-changing radiation amplitude. A useful leading parametrization is

$$k_{\text{in}}(\omega_n, \rho_n)R_{\text{tw}}(\omega_n) \simeq \pi[n + \theta(\epsilon_n)], \quad \rho_n \simeq \omega_n + c(\epsilon_n). \quad (10)$$

The radius and interior dispersion follow from the model approximations; the wall phase θ and dressed level shift contained in c remain calibrated. A separate coupled planar-wall calculation now supplies a numerical transmission zero, but does not determine both finite-curvature functions independently. Consequently, agreement of the calibrated phase rule with the sequence is not a parameter-free prediction.

For $\beta = 1/2$ and $\omega^2 = 1/2$, the exact planar profile is $S(x) = [1 + \exp(\sqrt{2}x)]^{-1}$. Solving the two-channel scattering problem, with one open and one evanescent channel on each side, gives

$$\rho_z \simeq 1.52414976213, \quad k_{\text{in}}(\rho_z) \simeq 1.92332459129. \quad (11)$$

An independently implemented Numerov boundary-value calculation confirmed this zero before reading the comparison calculation. It used a continuously phase-normalized transmission amplitude, actual complex-transmission residuals, flux and reciprocity controls, and separate mesh and boundary refinements. A conservative empirical frequency uncertainty is 10^{-9} ; this is not a certified enclosure. Two finite frequency scans found no additional zero, but do not prove uniqueness throughout the open-channel window. The planar scattering state has an incident and reflected wave; it is not itself a square-integrable embedded mode.

If $k_{\text{in}} \rightarrow k_\infty > 0$ and the wall phase has a finite limit, Equation (10) predicts, within the approximation,

$$\epsilon_n \sim \frac{1}{b_\infty n}, \quad b_\infty = \frac{2\sqrt{\beta}\pi}{k_\infty}. \quad (12)$$

From the rounded locations in Table 1, the inverse-detuning steps for indices 12–15 are approximately 2.3043, 2.3053 and 2.3068; these descriptive values do not specify accuracy beyond the input locations. The calibrated model gives 2.298, while its bare wall-state estimate gives 2.334. Substituting the independently computed planar zero into Equation (12) instead gives $b_\infty^{\text{wall}} \simeq 2.31000162857$, with a conservative empirical numerical uncertainty of 2×10^{-9} . This wall-based candidate was not fitted to ladder locations. Identifying it with the actual radial limiting step still requires a controlled curvature and phase-matching limit. This supports a useful spacing description; it does not prove existence of infinitely many localized modes or identify an

exact limiting constant. The phase model also motivates alternating signs of a matching Jacobian and quadratic linewidth minima, but neither property independently replaces a tail-controlled existence proof. The inverse-first-power scaling follows from the calibrated model, not from an independently established asymptotic exponent. Exploratory free power fits yielded exponents between -0.76 and -0.88 ; the reciprocal fit improved the comparison metric only by factors of approximately 1.2–2, below the originally sought factor of ten. Finite-data agreement therefore does not establish an exact asymptotic law.

A local numerical check also supports a planar transmission cancellation for the modified potential $U(S) = S - S^2 + S^3$ at $\omega_{\min} = \sqrt{3}/2$. An independent Numerov implementation, with the candidate frequency already known, gives $\rho_z \simeq 1.77345307180$ and $k_{\text{in}} \simeq 2.3994321234$. The last grid change, $h = 0.04$ to 0.02 , shifts ρ_z by 1.64×10^{-10} , while increasing the half-domain from 32 to 40 at the finest spacing changes it by 4.2×10^{-14} ; these are empirical sensitivities, not certified bounds. The corresponding thin-wall spacing candidate is $2\pi/k_{\text{in}} \simeq 2.618613482$. This known-target local replication neither counts all planar zeros nor establishes the radial ladder limit for the modified potential.

An exploratory two-dimensional follow-up found further numerical candidates near $\omega^2 = 0.529265$ and 0.525780 after refining the frequency sampling. The original decision rule remained inconclusive; the resolved numerical winding for the former candidate and the sign-change/linewidth evidence for the latter came from adaptively specified follow-up runs. The last estimated inverse-detuning steps are about 4.62, compared with about 2.305 in the three-dimensional sequence. This is not a dimensional scaling theorem. A similar sampling limitation in the earlier three-dimensional search remains a hypothesis.

6 Computer-assisted existence results

We distinguish the three computer-assisted existence results below from the numerically observed ladder. The proof establishes a localized mode at a parameter pair inside each specified box. It does not certify every observed radiation minimum, an infinite ladder, or a nonlinear nonradiating solution.

6.1 Model and precise statement

Consider the complex Klein–Gordon equation in three spatial dimensions,

$$\partial_t^2 \phi - \Delta \phi + U'(|\phi|^2) \phi = 0, \quad U(S) = S - S^2 + \frac{1}{2} S^3. \quad (13)$$

A stationary Q-ball has the form $\phi_0 = e^{i\omega t} f(r)$, where

$$f'' + \frac{2}{r} f' = (1 - \omega^2) f - 2f^3 + \frac{3}{2} f^5, \quad f(0) = a, \quad f'(0) = 0. \quad (14)$$

We use a real orthonormal basis $Y_{\ell m}$ of spherical harmonics and write a linear perturbation as

$$\delta \phi = e^{i\omega t} \left[\frac{A(r)}{r} e^{i\rho t} + \frac{B(r)}{r} e^{-i\rho t} \right] Y_{\ell m}. \quad (15)$$

In a complex harmonic basis the second sideband instead contains $Y_{\ell m}^*$. With $S = f^2$, substitution in the linearization of (13) gives

$$\begin{aligned} A'' &= \left[1 - 4S + \frac{9}{2} S^2 - (\omega + \rho)^2 + \frac{\ell(\ell+1)}{r^2} \right] A + CB, \\ B'' &= \left[1 - 4S + \frac{9}{2} S^2 - (\omega - \rho)^2 + \frac{\ell(\ell+1)}{r^2} \right] B + CA, \quad C = -2S + 3S^2. \end{aligned} \quad (16)$$

Regularity requires $A, B = O(r^{\ell+1})$. Define

$$\kappa_0 = \sqrt{1 - \omega^2}, \quad k = \sqrt{(\omega + \rho)^2 - 1}, \quad \kappa_c = \sqrt{1 - (\omega - \rho)^2}. \quad (17)$$

Computer-assisted existence statement. For each row of Table 3, the corresponding certificate encloses a quadruple $(a_*, c_*, \rho_*, \omega_*)$ with $c_* > 0$ such that (14) has a positive global solution satisfying $re^{\kappa_0 r} f(r) \rightarrow c_*$, and (16) has a nonzero real solution regular at the origin with

$$|A(r)| + |B(r)| \leq K_J e^{-\kappa_c r} \quad (r \geq L), \quad e^{\kappa_c r} B(r) \rightarrow 1. \quad (18)$$

The resulting spatial fields in (15) are smooth and exponentially localized. The certification is conditional on the correctness of the stated interval-arithmetic implementation and its underlying arithmetic libraries.

Case	ℓ	Enclosure of ω_*^2	Enclosure of ρ_*
C0	0	$0.797676787111865191750 \pm 4.8 \cdot 10^{-22}$	$1.744617544837398735781 \pm 6.8 \cdot 10^{-22}$
C1	0	$0.6851289044582160933 \pm 2.5 \cdot 10^{-20}$	$1.6903565973281434568 \pm 4.7 \cdot 10^{-20}$
C2	1	$0.7544960137826372331 \pm 7.8 \cdot 10^{-20}$	$1.826342030181069956 \pm 3.0 \cdot 10^{-19}$

Table 3: The three certified parameter boxes, displayed as outward-rounded decimal enclosures. The proof uses the full exact dyadic boxes in (a, c, ρ, ω) , not these two-coordinate summaries. C0 is BEWEIS-1; C1 and C2 are respectively T1 and T2 of BEWEIS-2. Their numerical ladder assignments are $(\ell, n) = (0, 1), (0, 2), (1, 1)$; the labels n are not separately certified nodal counts.

For the enclosed parameter pair and its fixed radial angular channel, every solution of (16) that is square integrable on $[L, \infty)$ is a scalar multiple of the constructed solution. This is *geometric radial simplicity*. It neither establishes algebraic simplicity of a dynamical generator nor excludes Jordan chains. For C2, the three-dimensional $\ell = 1$ angular multiplet remains degenerate. The translation mode at $\rho = 0$ does not belong to the fixed nonzero frequency considered here.

The term *embedded* is used operationally: $k^2 > 0$ and $\kappa_c^2 > 0$, so a free asymptotic channel is open, while the constructed mode is exponentially localized and has no oscillatory radiation tail. We do not identify the essential spectrum of a fully specified operator pencil in this proof. The perturbation is periodic after removing $e^{i\omega t}$; the two laboratory frequencies are $\omega \pm \rho$, with no claim about their rational commensurability.

6.2 Matching to an infinite exterior

The matching radius L is not a truncation at which the potential is discarded. Both the profile and the mode are continued to infinity by Volterra constructions. For the profile set $\tilde{u}(r) = re^{\kappa_0 r} f(r)$ and $q_f = -2f^2 + \frac{3}{2}f^4$. Its tail satisfies

$$\tilde{u}(r) = c + \int_r^\infty \frac{1 - e^{-2\kappa_0(s-r)}}{2\kappa_0} q_f(s) \tilde{u}(s) ds. \quad (19)$$

For a chosen bound $\bar{f} > 0$, put

$$K_P = \frac{(2 + \frac{3}{2}\bar{f}^2)e^{-2\kappa_0 L}}{4\kappa_0^2 L^2}, \quad \eta = 2K_P c^3, \quad m = c + \eta, \quad (20)$$

$$\Lambda = \frac{(6 + \frac{15}{2}\bar{f}^2)m^2 e^{-2\kappa_0 L}}{4\kappa_0^2 L^2}.$$

The conditions $me^{-\kappa_0 L}/L \leq \bar{f}$, $K_P m^3 \leq \eta$, $\Lambda < 1$, and $c > \eta$ give a positive unique fixed point in $\|\tilde{u} - c\|_\infty \leq \eta$. Common interval majorants verify these inequalities over the entire parameter box. They also bound the corrections to both profile matching data.

For the mode, write $dV = -4f^2 + \frac{9}{2}f^4$ and use $\tilde{A} = e^{\kappa_c r} A$, $\tilde{B} = e^{\kappa_c r} B$. In the $\ell = 0$ case the normalized Volterra kernels are

$$K_A(r, s) = e^{-\kappa_c(s-r)} \frac{\sin(k(s-r))}{k}, \quad K_B(r, s) = \frac{1 - e^{-2\kappa_c(s-r)}}{2\kappa_c}. \quad (21)$$

The inhomogeneous free term is $(0, 1)$. For $\ell = 1$ use

$$\begin{aligned} j(x) &= \sin(x)/x - \cos(x), & y(x) &= -\cos(x)/x - \sin(x), \\ d(u) &= e^{-u}(1 + 1/u), & g(u) &= e^u(1 - 1/u), \\ G_A(r, s) &= \frac{j(kr)y(ks) - y(kr)j(ks)}{k}, \\ G_B(r, s) &= \frac{d(\kappa_c r)g(\kappa_c s) - g(\kappa_c r)d(\kappa_c s)}{2\kappa_c}. \end{aligned} \quad (22)$$

Here $K_{A,B} = e^{-\kappa_c(s-r)} G_{A,B}$ and the free normalized term is $(0, 1 + 1/(\kappa_c r))$. For $kL, \kappa_c L \geq 1$, valid bounds are

$$\begin{aligned} |K_A| &\leq a_1 = \frac{1 + (kL)^{-2}}{k}, & |K_B| &\leq b_1 = \frac{b_0}{2\kappa_c}, \quad b_0 = 1 + (\kappa_c L)^{-1}, \\ |e^{-\kappa_c(s-r)} \partial_r G_A| &\leq a_2 = \sqrt{1 + (kL)^{-2}}, & |(\partial_r - \kappa_c) K_B| &\leq b_2 = 1 + (\kappa_c L)^{-1} + \frac{1}{2}(\kappa_c L)^{-2}. \end{aligned} \quad (23)$$

In particular, the last bound concerns the normalized *physical* derivative, not $\partial_r K_B$ alone. Since $|dV| + |C| \leq 6f^2 + \frac{15}{2}f^4$ is integrable, ordered Volterra iteration converges with factorial majorants. With $Q \geq \int_L^\infty (|dV| + |C|) ds$ and $\mu = \max(a_1, b_1)$, the $\ell = 1$ correction bounds are $(a_1, b_1, a_2, b_2)\varepsilon$, where $\varepsilon = b_0(e^{\mu Q} - 1)/\mu$. Uniform convergence also supplies the parameter continuity needed below.

6.3 Finite-dimensional existence certificate

The four unknowns are $z = (a, c, \rho, \omega)$. Rigorous forward integration computes the profile and two regular radial solutions R_1, R_2 from the origin. For $\ell = 1$ the initial variables are $A = r^2\alpha$, $B = r^2\beta$; their regular equations have derivative term $4\alpha'/r$ or $4\beta'/r$. For a source F , the origin integral equation has weight $r^2 \int_0^1 \tau(1 - \tau^3)F(\tau r) d\tau/3$, of mass $r^2/10$. This avoids starting a singular $2/r^2$ equation with approximate data. Subsequent interval Taylor steps use checked complex a priori enclosures and Cauchy remainder bounds; parameter variations enclose the required Jacobian over the whole box.

The matching map is $H = H_0 + E$. Its first two rows match profile value and derivative. The last two match the Wronski forms $W(\Psi, R_i) = \Psi \cdot R'_i - \Psi' \cdot R_i$, using free Jost data in H_0 and bounded tail corrections in E . For $\ell = 0$, these two Wronskis equal the origin values of Ψ . For $\ell = 1$, regular solutions behave as $r^2 e_i$ and normalized singular solutions as $r^{-1} e_i/3$; their Wronski pairing is δ_{ij} . Vanishing of both Wronskis therefore removes both singular coefficients.

Let $Z = z_0 + [-\delta, \delta]$, let D enclose $DH_0(Z)$, and suppose $|E_i| \leq \epsilon_i$ on Z . A fixed dyadic preconditioner Y is used to verify

$$z_0 - YH_0(z_0) + (I - YD)[- \delta, \delta] + |Y|[-\epsilon, \epsilon] \subset \text{int } Z, \quad \max_i \sum_j |I - YD|_{ij} \frac{\delta_j}{\delta_i} < 1. \quad (24)$$

The continuous map $z \mapsto z - YH(z)$ maps Z into itself, so Brouwer's theorem gives a fixed point. The weighted matrix bound makes Y invertible, hence the fixed point satisfies $H(z_*) = 0$. Differentiability of E is not required, and uniqueness of this parameter quadruple is not asserted. Profile positivity follows from positive interior enclosures, a minimum-principle argument on the remaining finite interval, and the positive exterior fixed point.

Finally, $2\kappa_0 > \kappa_c$, verified for all three boxes, permits construction of two bounded oscillatory comparison solutions at infinity. Their mutual Wronskian is nonzero. Together with the decaying Jost solution they span a three-dimensional subspace of the four-dimensional radial Cauchy-data space. Every square-integrable solution is Wronski-orthogonal to this subspace, whose symplectic complement is exactly the decaying line. This proves the geometric simplicity stated above.

6.4 Scope of verification

The defining certificates use $L = 40$ for C0, $L = 32$ for C1, and $L = 36$ for C2, with 256-bit ball arithmetic. Additional runs use larger matching radii and 320-bit arithmetic. C2's originally specified repeat at $L = 40$ failed the inclusion test. A subsequent run with an enlarged c -radius passed after reevaluating the bounds on the new box; this is an adaptive additional certificate, not an originally successful repeat. The C2 existence conclusion rests on its first successful certificate alone.

Project-internal cross-house mathematical and code reviews found no blocking error in the reviewed chains; the stated scope and harmonic conventions include their corrections. Review and higher-precision reruns do not constitute a second independent implementation. Correctness of Python-FLINT, Arb/Acb, and the recorded execution remains part of the computational trust base. For C0 the simplicity inequality was recorded and separately checked, but was not included in the historical aggregate success flag; the newer C1/C2 drivers include it explicitly. Source and input versions are tied to the archived certificates by hashes.

The validated arithmetic uses Arb/Acb midpoint-radius enclosures [14], accessed through Python-FLINT [15, 16]. The finite-dimensional test is a Krawczyk-type preconditioned inclusion [17]; for the separately bounded continuous tail correction, existence follows from the explicit Brouwer argument in Section 6.3, without asserting uniqueness of the parameter quadruple. These references identify the methodological background; the integration, tail bounds and inclusion hypotheses remain the specific checks of the archived certificates.

These results establish three linear localized modes in specified parameter boxes. They do not establish nonlinear radiation cancellation, linear or nonlinear stability, algebraic spectral simplicity, the absence of other modes or parameter zeros, or the existence of every member of the empirical ladder.

7 Nonlinear sources and the limits of linear trapping

A localized eigenmode of the linearized equation need not remain radiationless at finite amplitude. We distinguish an exact source calculation, the kinematic opening of radiation channels, and a nonzero outgoing response. Existing nonlinear evidence for radial location C0 is described below; it is not transferred to the additional modes T1 and T2 without evaluating their own sources.

7.1 Quadratic expansion

We use the complex nonlinear Klein-Gordon equation

$$\phi_{tt} - \Delta\phi + U'(|\phi|^2)\phi = 0, \quad U(S) = S - S^2 + \frac{1}{2}S^3, \quad \phi = e^{i\omega t}(f + z), \quad (25)$$

where the real background satisfies $-\Delta f - \omega^2 f + U'(f^2)f = 0$. Setting $s = f^2$, the linear operator is

$$\mathcal{L}z = z_{tt} + 2i\omega z_t - \Delta z + [U'(s) + sU''(s) - \omega^2]z + sU''(s)\bar{z}. \quad (26)$$

The quadratic term in the equation's left-hand side is exactly

$$\begin{aligned} G_2[z] &= fU''(s)(z^2 + 2|z|^2) + \frac{1}{2}f^3U'''(s)(z + \bar{z})^2 \\ &= \mathcal{A}z^2 + 2\mathcal{A}|z|^2 + \mathcal{C}\bar{z}^2, \end{aligned} \quad \mathcal{A} = f\left(-2 + \frac{9}{2}f^2\right), \quad \mathcal{C} = \frac{3}{2}f^3. \quad (27)$$

Consequently $z = \epsilon z_1 + \epsilon^2 z_2 + \dots$ gives $\mathcal{L}z_1 = 0$ and $\mathcal{L}z_2 = -G_2[z_1]$. In particular, the forcing has the opposite sign to the Taylor term in Eq. (27).

For a real spherical harmonic Y and real radial functions a, b , write

$$z_1 = Y(ae^{i\rho t} + be^{-i\rho t}). \quad (28)$$

Then

$$G_2[z_1] = Y^2(Q_+e^{2i\rho t} + Q_0 + Q_-e^{-2i\rho t}), \quad (29)$$

$$Q_+ = \mathcal{A}(a^2 + 2ab) + \mathcal{C}b^2, \quad Q_- = \mathcal{A}(b^2 + 2ab) + \mathcal{C}a^2, \quad (30)$$

$$Q_0 = 2\mathcal{A}(a^2 + b^2) + 2(\mathcal{A} + \mathcal{C})ab. \quad (31)$$

These expressions introduce rotating-frame frequencies $0, \pm 2\rho$, or laboratory frequencies $\omega, \omega \pm 2\rho$. The zero-frequency term is static only in the rotating frame.

7.2 Angular selection and forced channel problems

For T1, the radial $l = 0$ mode, $Y = Y_{00}$ and $Y_{00}^2 = Y_{00}/\sqrt{4\pi}$, so only $L = 0$ occurs at this order. For T2, choose a real normalized $l = 1$ harmonic. After a spatial rotation it equals Y_{10} , and

$$Y_{10}^2 = \frac{Y_{00}}{\sqrt{4\pi}} + \frac{Y_{20}}{\sqrt{5\pi}}. \quad (32)$$

Thus each of the three temporal source coefficients can drive $L = 0$ and $L = 2$; the antisymmetric $L = 1$ product is absent. A different real orientation rotates the quadrupole within the $L = 2$ multiplet. This standing real mode must not be confused with a rotating complex $Y_{11}/\bar{Y}_{11}$ block: the latter has $Y_{11}^2 = \sqrt{3/(10\pi)}Y_{22}$ for its corresponding oscillatory product.

Let c_L denote the coefficient in the relevant angular product and $D_L = -\partial_r^2 - 2r^{-1}\partial_r + L(L+1)r^{-2}$. For the oscillatory response $h_{+,L}e^{2i\rho t} + h_{-,L}e^{-2i\rho t}$ one must solve the coupled system

$$\begin{pmatrix} D_L + U' + sU'' - (\omega + 2\rho)^2 & sU'' \\ sU'' & D_L + U' + sU'' - (\omega - 2\rho)^2 \end{pmatrix} \begin{pmatrix} h_{+,L} \\ h_{-,L} \end{pmatrix} = -c_L \begin{pmatrix} Q_+ \\ Q_- \end{pmatrix}. \quad (33)$$

Here U', U'' are evaluated at s ; the conjugated second unknown matters when imposing outgoing conditions. The real static correction obeys $[D_L + U' + 2sU'' - \omega^2]h_{0,L} = -c_L Q_0$. Gauge and charge constraints must also be fixed in constructing a nonlinear continuation.

Both T1 and T2 have $|\omega| < 1$ and $|\omega \pm 2\rho| > 1$. For T2, the latter frequencies are approximately 4.52130 and -2.78407 . The second harmonics are therefore open in the mass-one vacuum, whereas the rotating-static correction is closed. The centrifugal term does not change the asymptotic mass threshold. With the convention e^{iEt} , an outgoing wave has radial phase $e^{-i \operatorname{sgn}(E)\sqrt{E^2-1}r}$; a negative E does not mean negative outward energy transport.

An open channel and a nonzero local source do not establish radiation: the outgoing source overlap can vanish. A localized continuation without incoming radiation requires both open amplitudes to vanish in every allowed L block of Eq. (33). The T1 response remains open; an approximate-input T2 calculation is reported below. A nonzero such amplitude would obstruct a regular localized perturbative continuation with the specified leading mode, but would not exclude all other nonlinear states. Frequency shifts of order ϵ^2 do not automatically cancel the second harmonic: the carrier shift contributes a static background correction at this order, while a shift of ρ first acts on ϵz_1 at order ϵ^3 .

7.3 Complex dipole polarization and a conditional obstruction

The threefold dipole degeneracy allows more than rotations of a real standing mode. Let $Y_i(\mathbf{n}) = \sqrt{3/(4\pi)}n_i$ be real orthonormal Cartesian dipole harmonics and write $A(\mathbf{n}) = \sum_i p_i Y_i(\mathbf{n})$, with

$\mathbf{p} \in \mathbb{C}^3$. For the same real radial eigenmode as above, the allowed tangent has the linked sidebands $z_1 = a(r)Ae^{i\rho t} + b(r)\bar{A}e^{-i\rho t}$. This does not allow independently chosen polarizations of the two sidebands or a mixture of distinct radial modes.

Set $\nu = \mathbf{p}^\dagger \mathbf{p}$ and $\chi = \mathbf{p}^T \mathbf{p}$. The oscillatory quadratic forcing is proportional to A^2 , rather than $|A|^2$. Its orthogonal angular projections satisfy the exact identities

$$\|P_0(A^2)\|_{L^2(S^2)}^2 = \frac{|\chi|^2}{4\pi}, \quad \|P_2(A^2)\|_{L^2(S^2)}^2 = \frac{3\nu^2 - |\chi|^2}{10\pi} \geq \frac{\nu^2}{5\pi}. \quad (34)$$

These follow by decomposing $\mathbf{p}\mathbf{p}^T$ into its trace and symmetric traceless parts and integrating the fourth spherical moment. Circular polarization, for example $\mathbf{p} = (1, i, 0)/\sqrt{2}$, has $\chi = 0$ and hence no oscillatory $L = 0$ source, but its $L = 2$ source does not vanish.

This source identity has a conditional implication for the outgoing response. Define $C_{\text{ref}} \in \mathbb{C}^2$ to be the *exact* radial $L = 2, m = 0$ outgoing coefficient for the normalized Y_{10} tangent, with the same radial normalization. Assuming the exact regular outgoing problem exists uniquely, spherical symmetry makes its radial operator independent of m . If $b_m = \langle Y_{2m}, A^2 \rangle$, linearity gives

$$C_{2m} = \sqrt{5\pi} b_m C_{\text{ref}}, \quad \sum_m \|C_{2m}\|^2 = \frac{3\nu^2 - |\chi|^2}{2} \|C_{\text{ref}}\|^2 \geq \nu^2 \|C_{\text{ref}}\|^2. \quad (35)$$

The same identity holds separately for either radial component. Conjugating the second computational channel to the physical sideband permutes conjugate spherical harmonics and preserves these norm sums. Thus an exact radial $L = 2$ nonzero certificate would cover every nonzero complex polarization in this one radial dipole eigenspace. The numerical coefficients reported below do not provide that certificate.

For clarity, the corresponding nonlinear obstruction requires a precise continuation class. Suppose a family is twice continuously differentiable in amplitude as a 2π -periodic rotating-frame profile in $C^2(\mathbb{T}; H_\gamma^4(\mathbb{R}^3))$, where $H_\gamma^4 = \{w : e^{\gamma\sqrt{1+|x|^2}} w \in H^4\}$. For the exact T2 background and tangent we may fix $\gamma = 1/10$: the inherited exterior decay rates are at least 0.495 and 0.287, respectively. Their radial equations propagate this decay to derivatives through order four; the regular origin expansions are even for the background and of the form r^2 times even functions for the reduced dipole components. Consequently the unreduced Cartesian fields are smooth at the origin and belong to this same weighted space. This membership statement does not assume that a nonlinear continuation exists or certify a numerical Sobolev norm. Allow smooth carrier and modulation frequency shifts and an additional static first tangent, but no other first-order temporal harmonics. A nonzero exact second-harmonic outgoing amplitude would contradict localization of the second amplitude derivative. At this order the frequency-shift terms are static or fundamental, so they cannot remove the second-harmonic source. This excludes only that uniformly localized smooth continuation class; it is not a dynamical instability or lifetime theorem.

7.4 Numerical second-harmonic response of the dipole mode

We evaluated the T2 forcing in both $L = 0$ and $L = 2$ using separately reconstructed numerical profiles near the certified point. The reduced real mode components obey $\int_0^\infty (u^2 + v^2) dr = 1$, including an approximate tail contribution, with $\int Y_{10}^2 d\Omega = 1$. The solver uses the complex conjugate of our positive-carrier convention: both computational unknowns have positive- q outgoing Hankel conditions, and the second unknown is conjugated to reconstruct the physical negative sideband. The power is invariant under this convention change.

Three prescribed settings use exterior radii 48, 64, 64, source spacings 0.02, 0.02, 0.01, and relative ODE tolerances 2×10^{-10} , 2×10^{-10} , 2×10^{-11} . Two further solves at the finest settings use the coarser reconstruction level from the same input algorithm, with different exterior radius and tolerance. This is an input-sensitivity comparison, not an independent implementation. All four outgoing coefficients remain nonzero under these changes; the largest observed relative complex-amplitude change is below 7.8×10^{-8} . This sensitivity is not a rigorous error bound.

At the finest setting the harmonically averaged power coefficients are

Angular sector	Positive sideband	Negative sideband
$L = 0$	2.4239×10^{-7}	6.0359×10^{-5}
$L = 2$	4.4956×10^{-7}	9.1587×10^{-5}

The leading power is ϵ^4 times the sum of these coefficients in the stated normalization. Both physical sidebands carry positive outward energy with the stated outgoing conditions. These values neither determine a lifetime nor measure a nonlinear time-evolution loss. They must not be compared directly with coefficients using the earlier radial maximum-amplitude normalization.

This calculation supports second-harmonic leakage for the approximate T2 input, beyond merely finding open channels and nonzero local sources. It does not certify leakage at the exact enclosed BIC. Float rounding places some input parameters outside the tiny certified box; sampled scaled Hermite-interpolant ODE defects have a sampled maximum of about 5.2012×10^{-6} (rounded upward to 5.3×10^{-6}). These are midpoint samples on 0.01-spaced grids extending to $r = 80, 96$, excluding $|r - 8| \leq 0.011$ around the splice, not uniform residual bounds. Differences between reconstruction levels are much smaller. Omitted exterior source and potential contributions were diagnosed, not converted into propagated amplitude-error bounds in that calculation. The separate validated monopole transfer in Section 7.5 addresses these errors using a fixed comparison function and the exact mode's Jost normalization; it does not certify the numerical power table.

A subsequent fixed-coefficient reconstruction checked the source-current identity for each angular sector. For the real symmetric forced operator, $J = \text{Im}(U^\dagger U')$ satisfies $J' = \text{Im}(F^\dagger U)$, where U denotes the two reduced response components and F their forcing. Both computational channels use positive-wavenumber outgoing waves. Keeping the saved matching coefficients fixed, an independent volume quadrature agrees with $\sum_j q_j |C_j|^2$ to relative residuals 9.6×10^{-11} for $L = 0$ and 1.3×10^{-11} for $L = 2$. This is a numerical consistency check of the finite forced problem, not an energy-power measurement or a validated error bound. It does not independently validate a shared source formula. The change between the two volume quadratures is 1.9×10^{-9} or smaller relative to the respective outgoing current; this observation is not a certified quadrature-error estimate.

7.5 A conditional enclosure for the monopole response

For the real dipole tangent at T2, we assembled a complete error budget for the $L = 0$ second-harmonic forced problem. Here L labels the radiated angular sector, whereas the underlying localized linear mode has $l = 1$. The calculation uses the exact-mode Jost normalization, with the evanescent reduced component satisfying $\lim_{r \rightarrow \infty} e^{\kappa_{ct} r} v(r) = 1$. It therefore has a different amplitude normalization from the numerical power table in Section 7.4.

The error identity includes the inner and outer source differences, operator differences, the residual of the fixed approximate response, its actual matching jumps and origin contribution, and the exterior boundary and coefficient-extraction errors. The exterior potential and source are included through infinity. No measured small linewidth or comparison between two numerical resolutions substitutes for these terms.

The following inherited hypotheses specify the exact problem to which the numerical error budget applies; they are not new conclusions of the scalar summation:

1. The T2 existence enclosure of Section 6 supplies one common exact parameter tuple $z_* = (a_*, c_*, \rho_*, \omega_*)$ in its certified dyadic box, a positive regular spherical profile f_* , and a nonzero real regular dipole mode. All profile, mode and parameter-error bounds refer to this same tuple. Neither uniqueness of the tuple nor the claim that every point of the box is an exact root is assumed.
2. At that exact tuple, the decaying fundamental Jost solution belongs to the span of the two regular origin solutions. This is the inherited regular/Jost identification from the T2

Wronskian conditions, not an inference from overlap of numerical residual intervals. Its reduced components are $O(r^2)$ at the origin and obey the same normalization $e^{\kappa_c r} v(r) \rightarrow 1$, with $\kappa_c = \sqrt{1 - (\omega_* - \rho_*)^2}$. The background and mode satisfy the analytic infinite-tail bounds: for $r \geq 36$ one may use $|f_*(r)| \leq (97/8)e^{-0.495r}/r$ and $|u(r)| + |v(r)| \leq 1.097e^{-0.287r}$.

3. The exact quadratic $L = 0$ source is the complete product source of the real standing dipole in Eqs. (27)–(33). The fixed nominal mode, source, response and comparison coefficient use the same amplitude convention: the nominal unit- L^2 mode is divided by the exact rational $s_0 = 2558188928468177/2251799813685248$, and quadratic quantities by s_0^2 . This does not identify s_0 with the norm of the exact mode. The computational carrier conjugation is retained throughout.
4. For the second harmonic, the exact reduced operator is $-\partial_r^2 - Q_*^2 + V_*(r)$ with real symmetric, locally bounded V_* and $Q_* = \text{diag}(q_+, q_-)$, $q_{\pm} = \sqrt{(\omega_* \pm 2\rho_*)^2 - 1}$. Both channels satisfy $5/2 < q_{\pm} < 23/5$. The exterior Volterra bounds hold at these exact frequencies, with the analytic potential and source tails retained through infinity. At $R = 64$, the exterior potential satisfies $\int_R^\infty \|V_*(r)\|_2 dr / (5/2) < 1/2$ and the source is integrable. Thus the exact outgoing reduction includes both its homogeneous maps and its inhomogeneous source corrections. Together with regular Dirichlet data at the origin, the positive outgoing matrix current gives the unique forced solution and adjoint extraction problem. This argument concerns two open second-harmonic channels, not the fundamental embedded mode.
5. The analytic and validated input bounds used in the error identity enclose the entire core and exterior, including exact-to-nominal frequency and map changes. In Euclidean vector and induced matrix norms, the inherited bounds $\int_0^\infty \|V_*(r)\|_2 dr < 25/4$ and the Volterra estimates give the adjoint amplitude weights $\|Z\|_2 < 26/5$ and $\|Z'\|_2 < 14$. Source and operator differences, response residuals, origin and matching jumps, exterior source terms, and coefficient-extraction rounding are all bounded in this same convention, without dropping a tail or counting one twice. The directed-arithmetic implementation and its interval libraries remain part of the computational trust assumptions.

Under these inherited hypotheses, the computed common coefficient-error bound is

$$\|C_* - C_{\text{num}}\|_2 \leq \mathcal{E} < 0.001023269269845280. \quad (36)$$

The frozen comparison coefficient in the second computational channel has the lower bound

$$|C_{\text{num},2}| > 0.001582591541472706. \quad (37)$$

The directed arithmetic establishes, before rounding for display,

$$|C_{*,2}| > 0.000559322271627427. \quad (38)$$

This channel has laboratory frequency $\omega_* - 2\rho_* < 0$. The solver uses a negative carrier and conjugates its second computational component to recover that physical sideband. Conjugating the entire field to the positive-carrier convention used here gives $C_{-, \text{paper}} = C_{*,2}$; all these operations preserve the modulus. The first channel remains undecided by this common error bound; that failure does not imply its coefficient vanishes.

This result supplies the nonzero-coefficient premise of the conditional smooth-continuation argument for a real standing dipole. Its extension to a complex dipole polarization is proportional to $\mathbf{p}^T \mathbf{p}$ and therefore does not cover circular polarization. It gives neither a nonlinear lifetime nor a dynamical instability theorem. In particular, the bound is not a certified value for any power coefficient in the preceding approximate-input table.

7.6 Existing evidence for radial location C0

Separate computations already address the earlier $n = 1$, $l = 0$ mode near $\omega^2 = 0.7976767871$ and $\rho = 1.7446175448$. They concern the same sextic field equation, not T1 or T2. A finite-amplitude radial evolution compared the nonlinear field with its background and linear response on two spatial grids. For amplitudes 0.01, 0.02, 0.04, the integrated quadratic flux of the nonlinear remainder increased by factors approximately 15.97 and 15.89 upon doubling amplitude. This supports fourth-order scaling over the tested range; that remainder flux is not an exact measurement of the total lost core energy.

A subsequent harmonic diagnostic identified the dominant laboratory sideband $E = \omega - 2\rho \simeq -2.59611$ as outgoing at two detector radii, on both grids and in both prescribed time windows. The weaker $E = \omega + 2\rho$ component was resolved but did not satisfy every outgoing-wave criterion. Large residuals in the full field's harmonic fit remained, so these observations do not establish a periodic multifrequency solution.

A separately solved frequency-domain problem, using the same background and mode data but no fitted amplitude from the time evolution, predicted the dominant outgoing coefficient. At $\epsilon = 0.02$, its predicted amplitude was 5.1563×10^{-5} , compared with measured values approximately $(5.03\text{--}5.10) \times 10^{-5}$. The prescribed comparison criteria passed. Shared input profiles make this a separate mechanism test, not a wholly independent reconstruction of the background.

An interval calculation further separated one necessary overlap from zero for an explicitly frozen finite operator on $[0, 44]$:

$$I_2 \in [-0.16383470526432667, -0.10547329373338812]. \quad (39)$$

The corresponding I_1 interval contained zero, and the original test requiring both exclusions remained failed. Nevertheless, one certified nonzero overlap suffices to violate the necessary simultaneous condition $I_1 = I_2 = 0$ for that frozen problem. Its binary64 spline inputs are treated as exact data, with a free exterior and zero source beyond radius 44. This is not a certificate that the exact continuum Q-ball and its BIC have that overlap: a validated link to the continuum inputs and exterior tail remains required. The linear existence certificates do not, by themselves, supply that quantitative link.

If a leading second-order outgoing amplitude is nonzero, its power is of order ϵ^4 . A lifetime law additionally requires a controlled relation to mode energy and slow amplitude evolution, and separation of initial adjustment from persistent radiation. Neither the finite-time observations nor the frozen-operator certificate provides a general nonlinear stability or lifetime theorem. Earlier real-field shell-model normal forms describe a different model and supply no substitute for these Q-ball source and overlap conditions.

8 Interpretation, limitations and next discriminating tests

The localized modes are excitations of an extended classical field background. Nothing in this calculation identifies them with quarks, spin-1/2 particles, gauge bosons, or dark matter. Angular degeneracy is not a derivation of a local internal gauge symmetry. Likewise, geometric simplicity of a radial solution space does not imply algebraic simplicity of a frequency eigenvalue or nonlinear orbital stability. The positive conserved form on a selected mode is neither a linear nor a nonlinear stability theorem for the background or its full perturbation space.

8.1 Environmental sensitivity and possible formation mechanisms

Equation (1) contains a scalar field with global U(1) symmetry, but no electromagnetic field or coupling to other particle species. Its nonzero tail does not by itself imply interaction with cosmic-microwave-background photons. Such a calculation would require an explicit coupling and a dimensional scale.

On a fixed background, the conditional mode-projection identity separates homogeneous scattering from prescribed local forcing: a homogeneous solution with initially zero projection cannot populate it when the Green boundary form vanishes, whereas a local source with nonzero overlap can. The assumptions, forcing comparisons and source-work diagnostics are retained in Appendix A. They are distinct from the linear existence enclosures and the nonlinear radiation obstruction above.

An additional radiation channel. A separate leading-order probe adds a neutral scalar through $\mathcal{L}_{\text{int}} = -g\chi|\phi|^2$. On three fixed radial profiles, the finite-grid form factor $I(k) = \int_0^\infty f(a+b)j_0(kr)r^2 dr$, $k = \sqrt{\rho^2 - m_\chi^2}$, had respectively one, one and two detected sign-changing zeros in the sampled interval $0 < k < \rho$. These are candidate Born-amplitude cancellations, not a complete count of zeros of the exact integral. The coupling already sources a static neutral field through $(-\Delta + m_\chi^2)\chi_0 = -gf^2$; thus the unchanged Q-ball with $\chi = 0$ is not a solution of the coupled theory. Background and mode backreaction must be included before claiming a coupled bound state. Closing this channel with a strict threshold gap removes propagation at this harmonic, but does not by itself preserve the old channel cancellation or prohibit radiation at higher harmonics.

The exploratory concentration tests use a different, nonstationary initial background. Appendix B retains the non-dilution bound, angular response, exterior diagnostic and constrained transport comparisons, including their preparation dependence and finite-time limits. None establishes formation of a stationary Q-ball or binding. These tests do not supply additional evidence for the certified BIC backgrounds. The appendix also records a later spherically symmetric relaxation study: two individual Gaussian starts passed its charge–frequency diagnostic, but its predeclared combined formation criterion was not met. That limited observation does not resolve the broader formation question.

8.2 Exploratory sensitivity to the numerical lattice

Exploratory two-dimensional calculations at the seventh radial location also indicate sensitivity to numerical spatial discretization. Here the width is the amplitude decay rate $\Gamma = -\text{Im } \rho$ for the convention $\exp(-i\rho t)$. The reported minimum widths scale approximately as h^4 for a five-point square stencil and h^8 for both a weighted nine-point square stencil and a seven-point triangular stencil. The leading angular radiation components are $\cos(4\theta)$ for both square stencils and $\cos(6\theta)$ for the triangular stencil. These observations are consistent with a perturbative interpretation in which the leading anisotropic operator correction induces a radiation amplitude linear in its strength, and the width is quadratic in that amplitude. The analogous quadratic asymmetry law is established for optical quasi-BICs by Koshelev et al. [18], Eqs. (2)–(3); its application here remains a model-specific interpretation. The reported exponent uncertainties are fit errors, and comparisons of absorbing layers and shifted eigenvalue solves are sensitivity diagnostics rather than certified error bounds. The calculations concern one two-dimensional mode and do not establish three-dimensional or nonlinear persistence. They motivate explicit stencil and resolution checks before angular radiation patterns are interpreted as physical structure.

8.3 Remaining mathematical and experimental tasks

There are three distinct remaining mathematical tasks. First, a full operator-theoretic realization would sharpen the operational use of “embedded” by specifying the generator, its domain and its essential spectrum. Second, an independent interval implementation would test the proof software beyond source review and parameter repetitions. Third, the additional radial case requires its nonlinear outgoing amplitudes, while the dipole requires a complete quadrupole certificate to address the circular polarizations not covered by the conditional monopole bound in Section 7.5.

Neither that calculation nor the earlier evidence for radial location C0 provides a general continuum theorem of finite-amplitude localization. A short long-lived trajectory cannot settle that question.

For the dipole calculation, the finite-problem source–current balance has been checked without refitting the boundary coefficients. Propagated bounds for the reconstructed input and full exterior now close the conditional monopole comparison. The resulting nonzero coefficient obstructs only the uniformly localized smooth continuation class stated in Section 7.3, under the inherited exact-mode and analytic hypotheses. An independently implemented enclosure and the unresolved quadrupole response remain separate next checks.

An experimental analogue would additionally need a dimensional mapping, an identified platform implementing the two-channel operator, measurable loss channels and a separation from instrumental linewidth floors. None is established here. The field model and all displayed parameters remain dimensionless; no laboratory frequency or length is inferred.

9 Conclusion

The strongest results are three specific computer-assisted linear existence enclosures with analytic exterior treatment, rather than a claim that every observed width minimum is an exact bound state. A larger numerical sequence and successive predictions support a thin-wall phase-matching description, while leaving an infinite-ladder theorem open. The nonlinear expansion identifies additional radiative channels; a numerical dipole calculation supports second-harmonic leakage for an approximate input, and a complete error budget establishes a conditional nonzero monopole coefficient at the enclosed dipole mode. This does not cover circular polarization or establish a lifetime. Linear cancellation, approximate spacing, and nonlinear persistence are separate questions throughout.

A Prescribed driving and response diagnostics

Linear coupling to the localized mode. For radiation of the same scalar field, a conditional projection identity makes the distinction between scattering and local forcing precise. In a fixed real $l = 1$ harmonic, complexify the doubled reduced perturbation $\psi = (rz_l, r\bar{z}_l)^T$ and write an additive source as S . Equation (26) and its conjugate give

$$\psi_{tt} + 2i\omega\sigma_3\psi_t + \mathcal{K}\psi = S, \quad \mathcal{K} = -\partial_r^2 + \frac{2}{r^2} + (D - \omega^2)I + C\sigma_1, \quad (40)$$

where D, C are those of Eqs. (4)–(5), $\sigma_3 = \text{diag}(1, -1)$, and $\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. The physical perturbation is recovered by the conjugate frequency pairing. For a physical additive source in this real harmonic block, $S_2 = \overline{S_1}$; independent components are used only in the formal complexification, with the conjugate pairing restored for physical forcing. Use the radial inner product with measure dr and the regular self-adjoint domain of $\mathcal{K}$. With $X = (\psi, \psi_t)^T$, define

$$\mathcal{G} = \begin{pmatrix} 0 & I \\ -\mathcal{K} & -2i\omega\sigma_3 \end{pmatrix}, \quad \mathcal{J} = \begin{pmatrix} 2\omega\sigma_3 & -iI \\ iI & 0 \end{pmatrix}. \quad (41)$$

Then $\mathcal{G}^\dagger \mathcal{J} + \mathcal{J} \mathcal{G} = 0$ on that domain. For a nonzero localized fundamental mode $H(\rho)u = 0$, where $H(\rho) = \mathcal{K} - \rho^2 I - 2\omega\rho\sigma_3$, put $X_u = (u, i\rho u)^T$. If $\rho > |\omega|$, its conserved form is positive:

$$N = \langle X_u, \mathcal{J}X_u \rangle = -\langle u, H'(\rho)u \rangle = 2\langle u, (\rho I + \omega\sigma_3)u \rangle > 0. \quad (42)$$

Here ρ is the fundamental frequency, not 2ρ . Positivity on this mode is neither a linear nor a nonlinear stability theorem for the full system. The corresponding left projection $c = \langle X_u, \mathcal{J}X \rangle / N$ obeys

$$\dot{c} = i\rho c - \frac{i}{N} \langle u, S \rangle, \quad (43)$$

provided the radial Green boundary form vanishes. On $[0, R]$ the additional term is $i[u^\dagger\psi - u^\dagger\psi']_0^R/N$. At the origin this follows from regular values $u, \psi = O(r^2)$ and regular derivatives $u', \psi' = O(r)$. At infinity the localized mode and its derivative must decay against the controlled scattering field and its derivative; this includes suitable finite wave packets.

Thus a homogeneous linear solution ($S = 0$) with $c(0) = 0$ cannot populate this projection on the fixed background. Spatial separation of an incoming packet alone does not imply exact orthogonality, because the localized mode has tails. A prescribed local additive source can instead have $\langle u, S \rangle \neq 0$ and drive the mode. For $c(0) = 0$ and a finite square-integrable projected source, Eq. (43) gives

$$N|c(T)|^2 \leq \frac{T}{N} \int_0^T |\langle u, S(t) \rangle|^2 dt. \quad (44)$$

This is a mode-projection bound, not a statement about the energy supplied by an unspecified environment. Although Eq. (42) is positive on this eigenspace, $\mathcal{J}$ is indefinite on the full phase space. No globally passive port model, loading rate, nonlinear formation mechanism, or indefinite maintenance follows. The local source and its physical coupling must be specified separately; the nonlinear radiation obstruction in Section 7 addresses a different question.

A prescribed local pulse. For the exact T2 mode, write $u = (A, B)^T$ with the reduced real components of Eq. (3), in the exact-mode Jost normalization specified in Section 7.5. The projection identity gives a concrete excitation test for this already existing bound mode. In dimensionless mass-one units, prescribe a source with real angular profile Y_{10} . Choose the radial envelope $\chi(r) = (r-1)^2(2-r)^2$ on $1 \leq r \leq 2$ and zero elsewhere, and temporal envelope $\eta(t) = \sin^2(\pi t)$ on $0 \leq t \leq 1$ and zero otherwise. The physical source in the original field equation is $J_\phi = \epsilon \chi Y_{10} \eta \exp[i(\omega + \rho)t]$, with real ϵ , the small strength of the external drive in this linear-response calculation. It is added to the right-hand side of Eq. (25). The reduced source in Eq. (40) is $S = \epsilon r \chi \eta (e^{i\rho t}, e^{-i\rho t})^T$; its conjugate channel is not independent and must be retained. For the fixed real radial mode components A, B , define

$$I_A = \int_1^2 r \chi(r) A(r) dr, \quad I_B = \int_1^2 r \chi(r) B(r) dr.$$

The resulting finite-pulse overlap is

$$L(0) = \frac{I_A}{2} + I_B \int_0^1 \eta(t) e^{-2i\rho t} dt.$$

With $N > 0$ from Eq. (42), vanishing initial projection and vanishing Green boundary form, the pulse leaves $c(1) = -i\epsilon e^{i\rho} L(0)/N$. Exact integration of the stored nominal polynomials, combined with their certified componentwise error bounds and the same certified frequency box, gives an enclosure separated from zero. Conditional on the same exact T2 mode identification and normalization, a conservative bound is $|L(0)| > 1/500$. This certifies nonzero linear mode projection for nonzero ϵ , vanishing initial projection and the stated boundary conditions; it does not determine an absolute excitation energy without the mode norm. Replacing Y_{10} by Y_{00} gives zero direct dipole projection by angular orthogonality; other modes are not excluded.

Finite detuning. The conclusion need not require an exactly tuned external source. Replacing its carrier by $\omega + \rho + \delta$ gives

$$L(\delta) = I_A \int_0^1 \eta(t) e^{i\delta t} dt + I_B \int_0^1 \eta(t) e^{-i(2\rho+\delta)t} dt.$$

Since $\int_0^1 t \eta(t) dt = 1/4$,

$$|L(\delta) - L(0)| \leq \frac{1}{4}(|I_A| + |I_B|)|\delta|.$$

The conservative bounds $|I_A| < 3/1000$, $|I_B| < 6/1000$ therefore imply $|L(\delta)| > 7/8000$ for $|\delta| \leq 1/2$. This is a sufficient excitation bound for the prescribed local source, not a linewidth or a decay rate. Its duration is dimensionless, not a measured physical loading time. Neither calculation describes formation of the underlying Q-ball, self-sustained excitation, or coupling to a specified cosmological field.

Canceling one excitation is not stopping the Q-ball. The source-projection identity also supplies a simple linear control prediction. Let $d = e^{-i\rho t}c$, initially zero, and apply two identical physical pulses of the preceding family, with the same local rotating-frame waveform and strength ϵ , separated by $\Delta \geq 1$. Their total projected drive is

$$I_{\text{pair}} = \epsilon L(\delta)(1 + e^{-i\rho\Delta}). \quad (45)$$

Consequently $\Delta = \pi/\rho > 1$ cancels the mode projection generated by the first pulse, even for a common local detuning δ . Both sideband terms in L are retained. This construction needs identical pulses in the rotating frame: in laboratory variables their sources obey $J_2(t) = e^{i\omega\Delta}J_1(t - \Delta)$. A sign-reversed second pulse at that same half-period instead doubles the rotating projected drive. For $|\delta| \leq 1/2$, the preceding bound ensures a nonzero first-pulse projection. Relative to that projection, a timing error h leaves the remainder $2|\sin(\rho h/2)| \leq \rho|h|$. These statements are analytic consequences of the fixed linear model. The following finite-time calculation tests the zero-detuning case only; it does not numerically establish the entire detuning or timing-error family. The common unknown modal norm cancels from the cancellation condition; extinguishing an arbitrary pre-existing excitation would require its complex amplitude and calibration. Other modes, continuum waves, radiation already emitted, and nonlinear backreaction are not canceled by this one condition. Neither zero modal projection nor zero linear radiation implies that the charged Q-ball itself has stopped rotating or disappeared.

A finite-time radial counterpulse test. We tested the two-pulse prediction by evolving the linearized complex radial $\ell = 1$ equation on the fixed nominal T2 background, with the conjugate physical channel retained. Four initially zero arms used no pulse, one pulse, two equal-sign local rotating-frame pulses separated by $\Delta = \pi/\rho$, and a sign-reversed second pulse at the same delay. The calculation used $R = 20$, $T = 6$, grid spacings $h = 0.04, 0.02$ and fourth-order Runge–Kutta time steps $h/8$, with forcing reported per unit linear drive amplitude. It is a radial linear PDE calculation, not a nonlinear three-dimensional evolution.

Both grids passed the predeclared projection-balance and comparison checks. The equal-sign pair reduced the final target projection below 2% of the single-pulse reference, while the sign-reversed pair agreed with twice the reference complex projection within 2%. These are numerical acceptance bounds, not interval-certified errors. The observed fine-grid ratio was approximately 3.61×10^{-5} (0.0036%); the coarse-grid ratio was approximately 1.44×10^{-4} . The balance includes the explicitly computed defect of the stored nominal mode on the spatial grid; the source-only discrepancy is reported separately. The global second-component defect norm, which includes the forced cutoff of the nominal mode at the finite boundary, actually increases on refinement; we do not claim global eigenfunction-error convergence. The short-time paired defect and projection checks concern the evolving, spatially confined response. Figure 5 also shows that the radial field norm need not vanish when the selected projection is suppressed. At $T = 6$ its fine-grid value was 0.0119332 for the equal-sign pair, compared with 0.00818567 for the single pulse. The mode-subtracted endpoint norm uses the nominal reconstruction $q - cA - \bar{c}B$ and is not an orthogonal energy decomposition. No total energy, charge removal, long-time stability or complete stillness of the charged background is inferred.

Changing the complete pulse shape. A subsequent fixed comparison used the same nominal linear radial $\ell = 1$ evolution, $R = 20$, $T = 6$ and $h = 0.04, 0.02$. It compared the annular drive

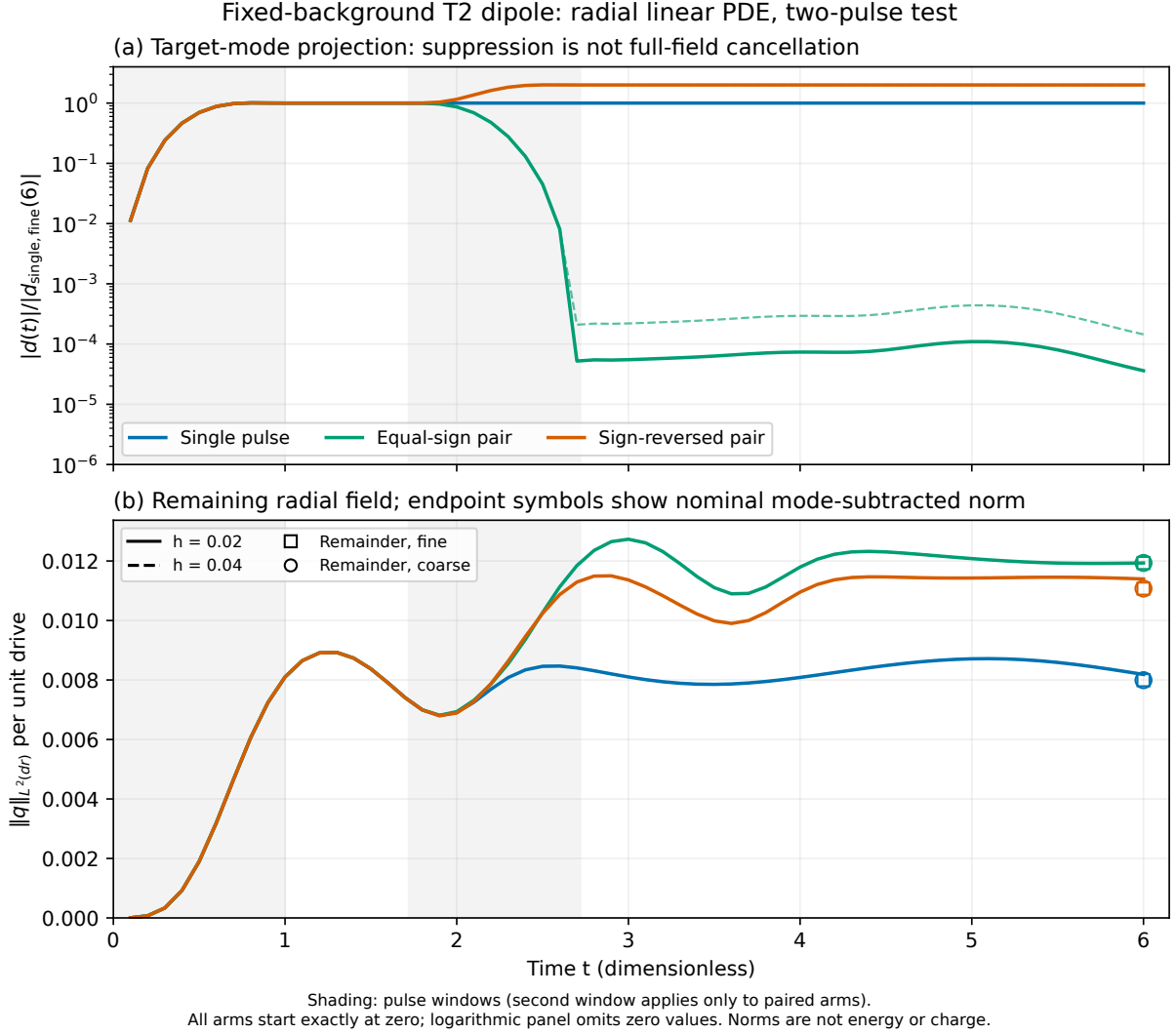


Figure 5: Stored radial linear-PDE histories for the T2 counterpulse test. Top: target projection magnitude, normalized by the fine-grid final single-pulse value. Bottom: reduced-field $L^2(dr)$ norms; endpoint symbols give norms after nominal mode subtraction. Colors identify pulse arms, solid/dashed lines the two grids. The exactly zero null arm is omitted from the logarithmic panel. Shading marks pulse support; only paired arms receive the second pulse. Lines connect recorded times without a new evolution. The laboratory second pulse includes the carrier-phase correction of the analytic construction. A small target projection coexists with a remaining field; these norms are neither radiated energy nor Noether charge.

above with a mode-adapted drive, using single pulses and equal-sign pairs separated by π/ρ . The new drive retains both sideband terms of a quintic turn-on construction and has a spatial cutoff equal to one for $r \leq 4$ and zero for $r \geq 8$. The reference instead uses the annulus $1 < r < 2$ and a $\sin^2$ temporal window with one rotating frequency. Thus both spatial and temporal waveforms change; the comparison does not isolate a spatial cause.

Each single pulse was normalized to the same reduced-source norm

$$B_{\text{src}} = \int \int |s(r, t)|^2 dr dt, \quad B_{\text{src}} \simeq 1.35281 \times 10^{-3} \quad (h = 0.02). \quad (46)$$

Pairs have twice this budget. The integrated budgets agreed with their prescribed values within 6×10^{-9} relatively on both grids. This norm is neither mechanical work nor injected energy or charge, and equal norms do not imply equal actuator support or hardware cost. For the single pulse, the fine-grid diagnostic $|d(T)|^2/B_{\text{src}}$ is about 32 times larger for the shaped drive (Table 4); this is not an energy-efficiency ratio or an optimality result.

Table 4: Fixed source-shaping comparison at $T = 6$, $h = 0.02$. Here $p = \partial_t q$, and “remainder” means subtraction of the nominal mode reconstruction. Field norms are $L^2(\text{dr})$ norms, not energies; the pair rows give the total remaining field and velocity norms.

Diagnostic	Annular drive	Shaped drive
Single-pulse $ d ^2/B_{\text{src}}$	0.001214	0.03872
Single-pulse q remainder norm	0.007992	0.002596
Single-pulse p remainder norm	0.01502	0.004954
Pair $\ q\ $	0.01193	0.001055
Pair $\ p\ $	0.02153	0.001930

The shaped pair leaves approximately 8.8% and 9.0% of the reference pair’s field and velocity norms, respectively; both remain nonzero. Nominal mode subtraction is not an orthogonal energy partition. Both grids passed the fixed comparison and defect-balance checks, but the global second-component nominal eigenmode residual norm increases from approximately 0.529 to 1.495 on refinement. The successful paired-defect balance therefore does not establish global eigenfunction convergence. These finite-box, short-time results show reduced residual motion for the specified composite drive; they establish neither full stillness of the field nor removal of the charged background, nonlinear stability or a lifetime. This original two-source comparison did not separate changes of spatial and temporal design. The subsequently specified cross-comparison below addresses that limitation within a fixed actuator family.

A subsequent stored-profile audit identifies the origin of the growing second-component defect in this box. The exported reference mode has $v(20) = 0.004228503237$, whereas the evolution operator imposes a zero Dirichlet value. For the three-point stencil this discrepancy adds $b_j = \delta_{j,n-1}v(20)/h^2$ to the continued-ghost residual, giving $\|b\|_h^2 = |v(20)|^2/h^3$. Its squared norm is 0.2793787442 and 2.2350299533 on the two grids; the continued residual squared norms are 1.52×10^{-8} and 9.50×10^{-10} , with signed cross terms -9.74×10^{-10} and -4.84×10^{-10} . The boundary discrepancy therefore dominates this component’s growing residual. It does not dominate the first component. This is an algebraic attribution, not a modification of the evolution boundary condition or a new eigenfunction calculation. The original global defect remains valid for the original operator. Two decreasing interior residuals neither certify the exported mode nor bound the integrated influence of the boundary discrepancy on the pulse response.

A fixed factorial actuator comparison. The new comparison uses two spatial pairs, $A = (w, 0)$ and $B = (\zeta u, \zeta v)$, where w is the preceding annulus profile, ζ the cutoff defined above (one below $r = 4$ and zero above $r = 8$), and u, v the stored nominal sidebands. Two temporal pairs are

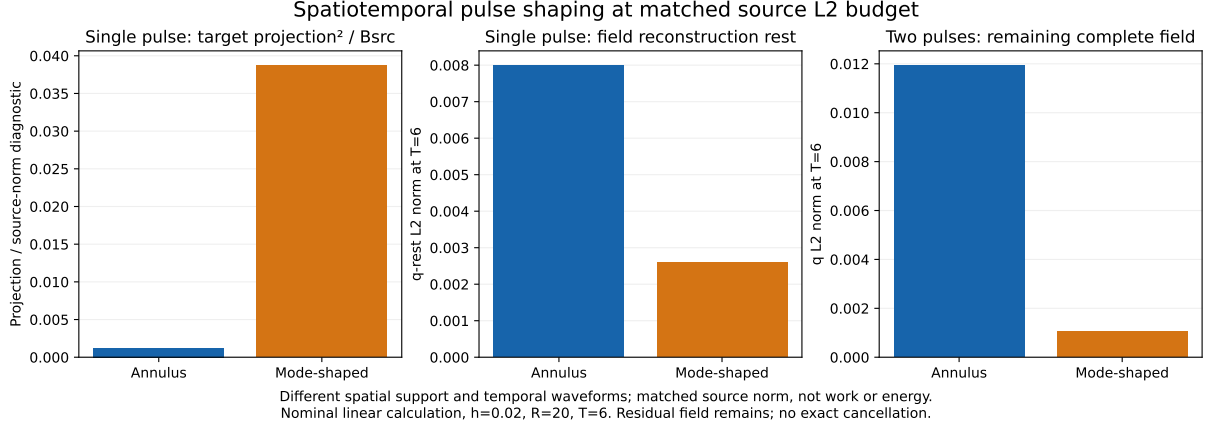


Figure 6: Endpoint diagnostics from the fixed spatiotemporal pulse comparison. Both spatial support and temporal waveform change. The common source norm is not energy or work. The shaped source increases the selected projection per source norm and reduces the residual response in this comparison; the complete field does not vanish. Bars show the stored fine-grid result, not a new evolution or a certified continuum limit.

prescribed on $0 < t < 1$:

$$T_0 = (\sin^2(\pi t)e^{i\rho t}, \sin^2(\pi t)e^{-i\rho t}), \quad (47)$$

$$T_1 = ([g'' + 2i(\rho + \omega)g']e^{i\rho t}, [g'' - 2i(\rho - \omega)g']e^{-i\rho t}), \quad g = 10t^3 - 15t^4 + 6t^5. \quad (48)$$

Both temporal pairs vanish outside $0 < t < 1$; g is extended constantly as zero before the pulse and one afterward. Each complex source is the component sum $s_{ij} = \mathcal{N}_{ij}(R_{i1}T_{j1} + R_{i2}T_{j2})$, where $R_i = A$ or B . There is no conjugation in this sum; the second physical equation receives s_{ij}^* . The positive factor $\mathcal{N}_{ij}$ fixes the same single-pulse B_{src} *after* summation, retaining component interference. The two old corners A/T_0 and B/T_1 are implementation controls; A/T_1 and B/T_0 are the new crossed drives. Each uses both a single pulse and an equal-sign pair at the unchanged delay π/ρ . The nominal operator, $R = 20$, $T = 6$, $h = 0.04, 0.02$ and time step $h/8$ remain fixed. Both grids passed the predeclared numerical checks.

Table 5: Four fixed normalized actuator designs at $T = 6$, $h = 0.02$. The projection diagnostic is for a single pulse; the final two columns are total remaining pair norms, not mode-subtracted norms or energies.

Design	$ d_{\text{single}} ^2/B_{\text{src}}$	Pair $\ q\ $	Pair $\ p\ $
A/T_0	0.001214	0.01193	0.02153
A/T_1	0.0004100	0.006390	0.01426
B/T_0	0.1586	0.02024	0.005692
B/T_1	0.03872	0.001055	0.001930

Within this family, B/T_0 gives the largest single-pulse projection diagnostic, whereas B/T_1 leaves the smallest pair norms in both q and p . Strong projected excitation and small residual field are thus different objectives. The spatial-pair change $A \rightarrow B$ increases the pair q norm at T_0 but decreases it at T_1 : its effect depends on the temporal partner. These are conditional design comparisons, not universal additive spatial and temporal effects. In particular A omits a spatial component, T_1 changes sideband weights and phase structure, and normalization depends on both factors. Equal source norm still implies neither equal work nor energy efficiency. All four pairs suppress their selected projection under the fixed criterion, while their full responses differ and remain nonzero. The nominal-mode defect and finite-box limitations above remain.

That fixed-phase comparison did not test phase-error robustness; it established neither exact preparation, complete stillness nor a lifetime.

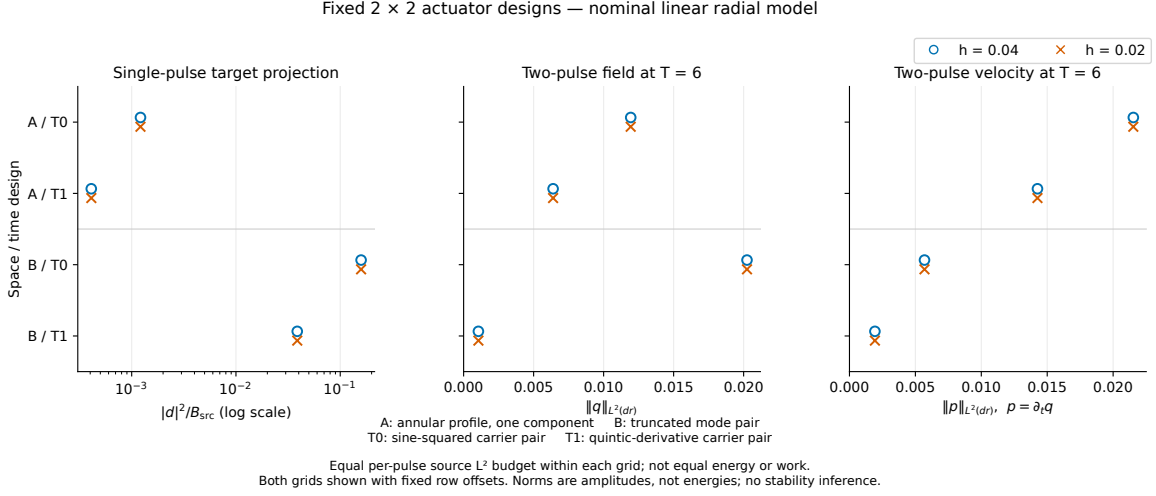


Figure 7: Stored endpoint diagnostics for the fixed 2×2 actuator family. The largest single-pulse projection per source norm occurs at B/T_0 , while B/T_1 gives the smallest remaining pair field and velocity norms. The source is normalized after adding its two components; these comparisons are not energy efficiencies or a universal separation of independent spatial and temporal mechanisms. No additional evolution or phase-error test is represented by this figure.

A constant relative pulse phase. A further bounded test changes only the additional phase θ of the second pulse, keeping the delay and the carrier-phase correction fixed. The coupling to q^* makes the nominal Bogoliubov evolution real-linear, not complex-linear: the response to is need not equal iq . For each of the four fixed designs, one new delayed- is response therefore completes the real basis. At the same endpoint,

$$q\theta = q_{\text{single}} + \cos \theta (q_{\text{pair},0} - q_{\text{single}}) + \sin \theta q_{\text{delayed},90}. \quad (49)$$

The same identity holds for p and the real-linear projection d . Four directly evolved $\theta = \pi/4$ pairs check this reconstruction: the largest reported relative discrepancy in q, p, d across both grids was 1.14×10^{-13} . This tests a known linear identity, not a new physical effect. Other constant phases below are evaluations of stored endfields, not additional time evolutions. The setting remains $R = 20$, $T = 6$, $h = 0.04, 0.02$.

Table 6: Selected constant-phase endpoints for B/T_1 at $T = 6$, $h = 0.02$. Norms are total remaining radial fields, not energy. The phase values were fixed before the endpoint analysis.

θ	$\ q\ $	$\ p\ $
0°	0.001055	0.001930
-5°	0.001062	0.002068
$+5^\circ$	0.001423	0.002462
-15°	0.001986	0.003720
$+15^\circ$	0.002562	0.004413
45°	0.006301	0.01125

These values need not be symmetric about zero phase. Cancellation of the selected coefficient and minimization of the full field norm are different conditions; no new minimum search was performed.

For a uniformly distributed phase that is *constant within each pulse pair*, write any endpoint field as $a + b \cos \theta + c \sin \theta$. Its root mean square norm is exactly

$$\sqrt{\mathbb{E}_\theta \|a + b \cos \theta + c \sin \theta\|^2} = \sqrt{\|a\|^2 + \frac{1}{2}(\|b\|^2 + \|c\|^2)}. \quad (50)$$

The Gram-matrix evaluation was also checked against eight equally spaced phases. Table 7 keeps all four designs in the comparison. Although B/T_1 leaves smaller pair norms than A/T_1 at zero

Table 7: Uniform constant-phase RMS endpoint norms, $T = 6$, $h = 0.02$. These are square roots of mean squared norms, not mean norms.

Design	$\sqrt{\mathbb{E}_\theta \ q\ ^2}$	$\sqrt{\mathbb{E}_\theta \ p\ ^2}$
A/T_0	0.01128	0.02168
A/T_1	0.007123	0.01447
B/T_0	0.02552	0.04088
B/T_1	0.01106	0.01954

phase, its two RMS norms are larger in this ensemble. Its fixed-phase advantage is thus conditional on phase preparation. The comparison fixes source norm, not the previously excited target amplitude: A/T_1 has a much smaller single-pulse projection diagnostic (0.0004100 versus 0.03872 for B/T_1). Its lower RMS therefore does not establish a better storage or erasure protocol at equal preparation. An RMS ranking is not necessarily an ordering of mean norms. This ensemble is neither a time average nor a simulation of fluctuating environmental noise. The constant phase does not change the disjoint pulses' source budget $2B_{\text{src}}$, but that budget remains distinct from work or energy. The global nominal eigenmode residual limitation remains unchanged; neither exact BIC preparation, complete stillness nor nonlinear or long-time robustness follows.

Quadratic endpoint energy and the rotating generator. A further analysis uses the stored single- and pair-pulse endpoints, without repeating the time evolution. With $p = q_t$ and the same h -weighted inner product and Dirichlet operator as the evolution, define

$$H_2 = \|p\|_h^2 + \langle q, Kq \rangle_h + \text{Re } h \sum_j C_j q_j^2, \quad (51)$$

$$Q_2 = 2 \text{Im} \langle q, p \rangle_h + 2\omega \|q\|_h^2, \quad E_2 = H_2 + \omega Q_2. \quad (52)$$

Here $K = -\Delta_h + 2/r^2 + 1 - 4f^2 + (9/2)f^4 - \omega^2$ and $C = -2f^2 + 3f^4$. There is no factor 1/2: H_2 is the coefficient of second order in $E - \omega Q$ for the stated action convention, using one physical complex field rather than counting its conjugate twice. This rotating quadratic generator need not be positive.

Table 8: Stored endpoint coefficients at $T = 6$, $h = 0.02$. All entries are in units of 10^{-4} , rounded to four significant digits. Within a fixed pulse count the designs have equal source-L2 budgets; a pair has twice the single-pulse budget.

Design	H_2 single	H_2 pair	E_2 single	E_2 pair
A/T_0	4.782	9.315	5.958	11.51
A/T_1	2.668	5.180	3.614	6.910
B/T_0	8.917	1.382	6.467	8.608
B/T_1	2.129	0.05203	1.201	0.03759

The distinction matters even for the direction of change: B/T_0 decreases H_2 while increasing E_2 from single to pair. Its Q_2 changes from approximately -2.821×10^{-4} to 8.319×10^{-4} . For

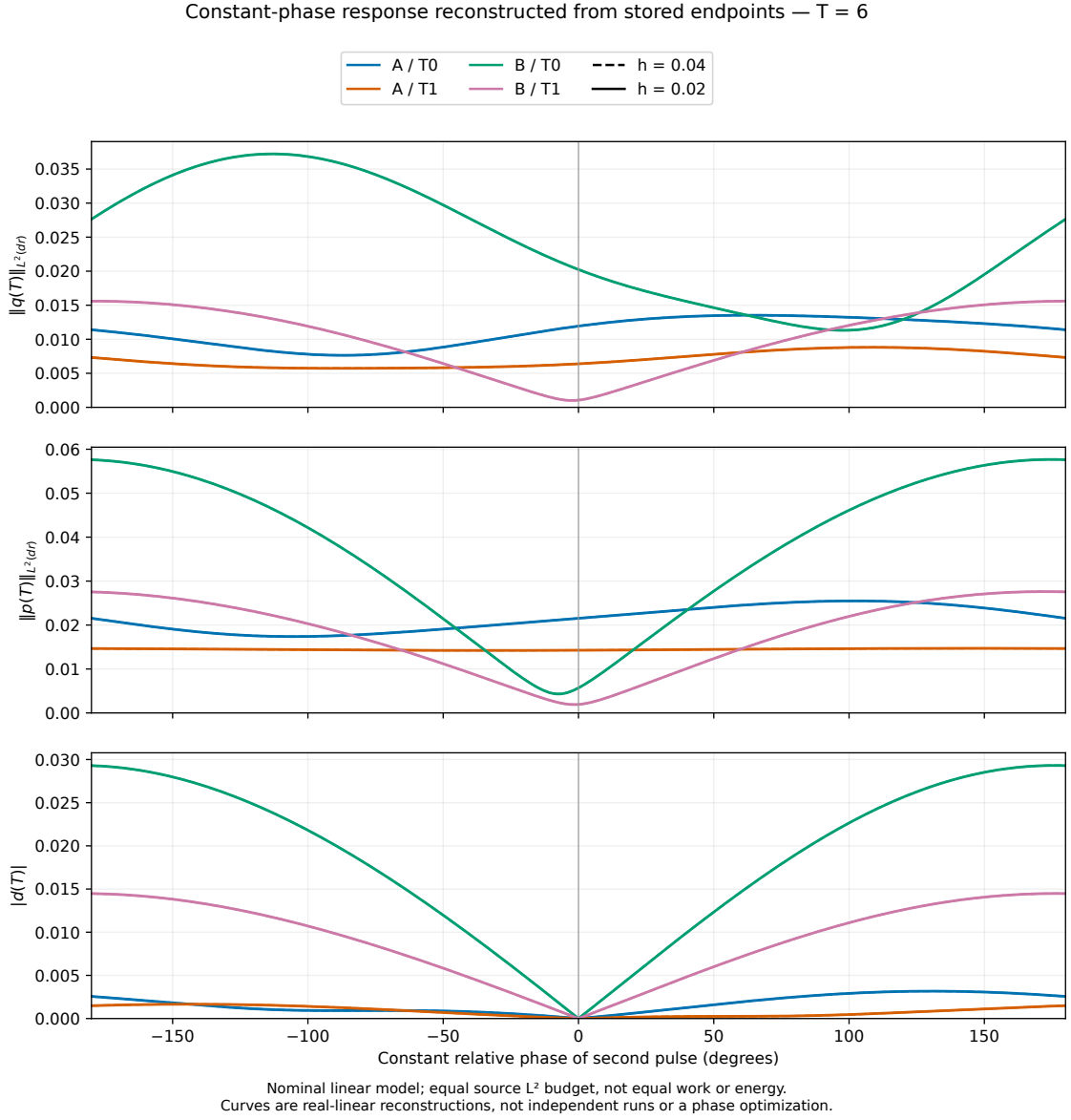


Figure 8: Constant-phase endpoint reconstruction for all four fixed actuator designs, from their stored real-linear response bases. The two grids compare nominal finite-box observables; plotted phase samples are not independent time evolutions. The additional phase is constant within a pulse pair, not a time-dependent noise source. Field norms and uniform-phase RMS values do not measure energy cost.

B/T_1 both endpoint coefficients decrease; for the two A designs both increase. These directions also occur on $h = 0.04$, but two-grid differences are not validated error bounds.

Equal source-L2 budgets do not imply equal work, and the single-to-pair comparison itself is not a same-budget comparison. For the fixed semidiscrete model, $\dot{H}_2 = 2 \operatorname{Re}\langle p, S \rangle_h$. Thus $H_2(T)$ at zero initial data predicts net rotating-generator work by an identity; that endpoint-only analysis did not measure an independent time-integrated power balance. The subsequent test below adds this observable. The quadratic laboratory source power additionally contains $2\omega \operatorname{Im}\langle q, S \rangle_h$, while $\dot{E}_2$ also contains the background-exchange term $2\omega \operatorname{Im} h \sum_j C_j q_j^2$. In particular, E_2 is a coefficient on the linear ansatz, not the complete laboratory energy change including a second-order background response. No positive or gross actuator work, energy efficiency, or full shutdown of the Q-ball is established by Table 8.

Time-integrated source work in the nominal linear model. A subsequent bounded run retains the same eight single/pair arms, $R = 20$, $T = 6$, $h = 0.04, 0.02$ and $\Delta t = h/8$, and adds power accumulators at the four Runge–Kutta stages. With the conventions of Eqs. (51)–(52), these integrate

$$P_H = 2 \operatorname{Re}\langle p, S \rangle_h, \quad J_Q = 2 \operatorname{Im}\langle q, S \rangle_h, \quad X_Q = 2 \operatorname{Im} h \sum_j C_j q_j^2, \quad (53)$$

$$P_{E,\text{src}} = P_H + \omega J_Q, \quad P_{E,\text{ex}} = \omega X_Q. \quad (54)$$

The independently evaluated state coefficients satisfy the numerical balances $\Delta H_2 = \int P_H dt$, $\Delta Q_2 = \int (J_Q + X_Q) dt$, and $\Delta E_2 = \int (P_{E,\text{src}} + P_{E,\text{ex}}) dt$. Unlike the preceding endpoint-only analysis, these integrals use the actual time-dependent fields and source at the integration stages. The exchange term remains active after the source ends.

The first pulse is identical in the single and pair arms; the first and second pulse windows are disjoint. Their trajectories therefore agree before the second pulse. Subtracting their independently integrated total source works gives the second pulse’s net work on the already excited field:

$$W_{H,2} = W_{H,\text{pair}} - W_{H,\text{single}}, \quad W_{E,\text{src},2} = W_{E,\text{src},\text{pair}} - W_{E,\text{src},\text{single}}. \quad (55)$$

This is not the work of an isolated second pulse applied to a fresh zero state, nor merely a relabeling of endpoint energy differences.

Table 9: Net second-pulse source work inferred from time-integrated pair-minus-single works at $h = 0.02$. Units are 10^{-4} , as in Table 8; entries are rounded to four significant digits. Negative values denote net extraction in the specified quadratic source-work convention, not efficiency of a physical actuator.

Design	$W_{H,2}$	$W_{E,\text{src},2}$
A/T_0	4.533	4.207
A/T_1	2.512	3.602
B/T_0	−7.535	5.939
B/T_1	−2.077	−1.122

Thus the second B/T_1 pulse is a net source-work sink in both conventions. In contrast, B/T_0 removes rotating-generator work while its laboratory source-work coefficient is positive. The change in E_2 also includes the difference of integrated exchange terms; it is not identical to $W_{E,\text{src},2}$. Positive and negative parts of the total spatially integrated power were separately accumulated, so a small net value need not mean little positive input or little extraction. This splitting applies after spatial summation, rather than to positive and negative local power densities.

Both grids pass the predeclared 10^{-6} normalized balance tolerance, and the new endfields agree with the stored reference endpoints within the fixed 10^{-8} single-response-normalized tolerance.

The reported scaled two-grid differences of work contributions are below 4.4×10^{-4} , within the prescribed 0.03 consistency limit. These are discrete numerical checks, not rigorous continuum error bounds; the positive/negative power split also need not inherit a smooth fourth-order convergence rate. The original nominal-mode boundary defect is unchanged. Equal source-L2 budgets are still not equal work. Finally, the laboratory source-work coefficient and E_2 belong to the frozen linear ansatz: second-order background response, full nonlinear laboratory energy, actuator losses and conversion efficiency are not determined. No complete shutdown or long-time stability follows from this short-window balance.

Phase dependence of the second-pulse source work. For the same four source designs, we vary the constant phase θ of the delayed pulse while keeping its source norm fixed. Real linearity of the frozen evolution gives the quadratic source-work coefficient

$$W_{E,\text{src},2}(\theta) = c_0 + c_1 \cos \theta + s_1 \sin \theta + c_2 \cos(2\theta) + s_2 \sin(2\theta). \quad (56)$$

Writing the measured quadrature coefficients as (A, B, D, E, F) , these are $(c_0, c_1, s_1, c_2, s_2) = ((D+E)/2, A, B, (D-E)/2, F/2)$. The five coefficients are integrated from the first-pulse response and two delayed quadrature responses, with a direct 45° two-pulse control, through $T = 3$ after both source windows end ($\Delta + 1 < 3$), using $h = 0.04, 0.02$ and $\Delta t = h/8$. This is not a renewed $T = 6$ energy history. Thus the plotted curves are postprocessing of measured coefficients, not thousands of independent evolution runs.

On both nominal grids, only B/T1 has a negative-work interval: approximately $-66.53^\circ < \theta < 57.03^\circ$. For the fine grid its second-pulse net source work is -1.1224×10^{-4} at 0° and -3.9278×10^{-5} in the direct 45° control. This broad nominal interval concerns signed net source work; it does not assert equally small residual fields throughout the interval. In particular, the previously reported residual field norms increase at 45° even though this work remains negative. All four uniform *constant-phase* means c_0 are positive; for B/T1 the fine-grid mean is 1.1334×10^{-4} . This phase ensemble is not a model of time-dependent phase noise.

Polynomial sign isolation treats the measured floating-point coefficients as exact rational dyads. Its exactness applies to those nominal polynomials, not to their discretization errors or to the continuum or nonlinear field equations. The result concerns quadratic source work in the frozen linear ansatz, with background exchange distinguished separately; it is neither a full nonlinear laboratory-energy balance nor an actuator-efficiency claim.

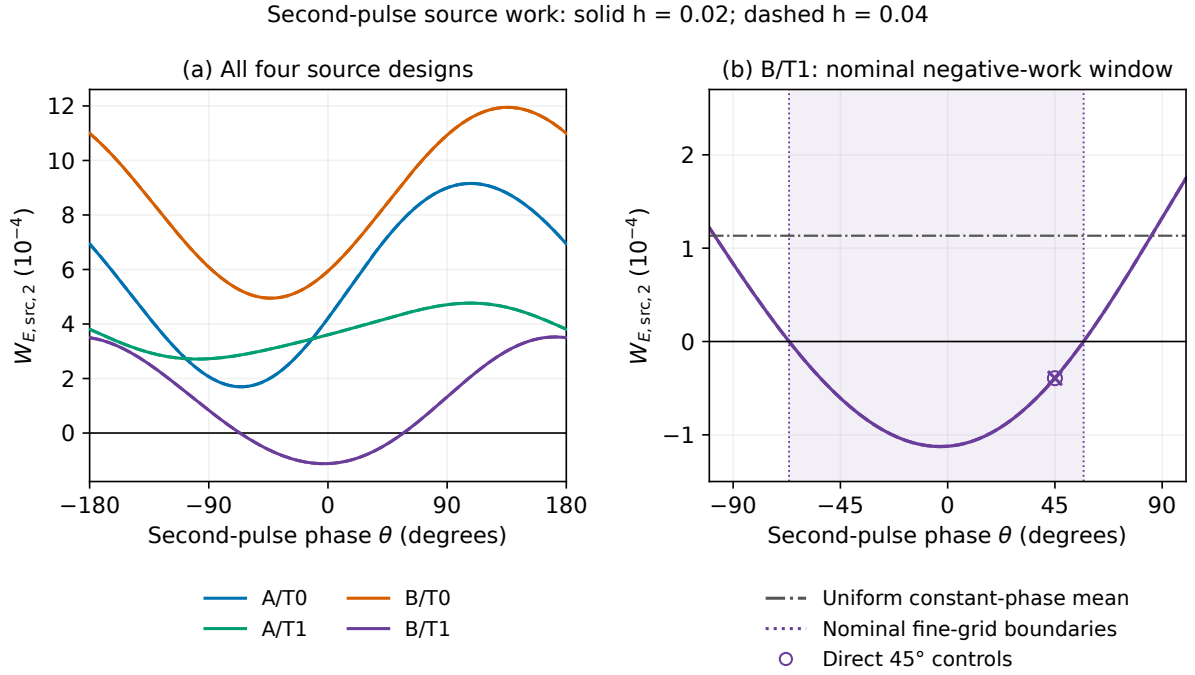


Figure 9: Second-pulse source work versus constant phase. Left: all four designs; solid curves use $h = 0.02$, dashed curves $h = 0.04$. Right: B/T1 with nominal fine-grid zero boundaries (dotted), its uniform constant-phase mean (dash-dotted), and direct 45° controls (symbols). Work is displayed in units of 10^{-4} . Shading denotes the nominal negative-work window, not certified physical tolerances.

One prescribed temporal phase drift. The constant-phase ensemble above does not determine the response to a phase varying during a pulse. As a separate finite comparison, we replaced the same delayed B/T1 source $S_d(t)$ by $S_{\delta,\theta_0}(t) = \exp[i\theta_0 + i\delta(t - \Delta)]S_d(t)$, with the single preselected value $\delta = 0.1$ against $\delta = 0$. The multiplier acts on the entire complex source; the original envelope, carrier construction and delay are unchanged. It preserves the source magnitude pointwise and its integrated L^2 budget. For each detuning we independently evolved the responses to $S_{\delta,0}$ and $iS_{\delta,0}$ through $T = 3$. Real linearity gives the response at θ_0 as the cosine-sine combination of these two responses; the second is not assumed to be i times the first.

Averaging uniformly over the *constant initial* phase θ_0 eliminates both the mixed quadrature term and the work cross term with any deterministic first-pulse field. Consequently the mean second-pulse laboratory source work is one half of the sum of the two isolated quadrature works. No first-pulse evolution was repeated; this reduction applies to the mean, not to phase-specific two-pulse work. With $(h, \Delta t) = (0.04, 0.005)$ and $(0.02, 0.0025)$, both means are positive on both discretizations. On the finer grid the mean increases from $1.133373236 \times 10^{-4}$ to $1.248632464 \times 10^{-4}$, a difference of $1.152592277 \times 10^{-5}$ (about 10.2%).

The increase exceeds the predeclared diagnostic comparison guard 7.94×10^{-11} , which combines the change between discretizations, unshifted-work reproduction errors and energy-balance residuals. This guard is not a certified error bound; space and time resolution change together. The result establishes sensitivity of this nominal mean source work to one prescribed phase drift at equal source norm. It does not give a phase-noise or thermal model, an efficiency gain, or a claim about all detunings. The frozen-profile approximation, artificial $R = 20$ boundary and omitted nonlinear backreaction remain unchanged.

Local direction and one mirror detuning. A follow-up fixed $d = 0.1$ and added only the mirror source at $-d$ and the parameter tangent at zero. Writing $z = \partial_\delta q|_0$, the exact differentiated finite-grid equation has the same real-linear operator and forcing $i(t - \Delta)S_d$. Its numerical integration retains both derivatives of the work integrand, from the response and from the source, for each independently evolved quadrature. The zero-detuning basis trajectories are needed for these products; the first pulse is not repeated, and the $+d$ result is reused. For the uniform initial-phase mean $m(\delta)$, the finer grid gives $m'(0) \simeq 1.1006065 \times 10^{-4}$ in model units. Both grids resolve a positive slope with the prescribed empirical guard. The three fine-grid means remain positive:

δ	Mean laboratory source work $m(\delta)$
-0.1	1.0285198×10^{-4}
0	1.1333732×10^{-4}
+0.1	1.2486325×10^{-4}

Define the finite odd and even changes by $O_d = [m(d) - m(-d)]/2$ and $E_d = [m(d) + m(-d) - 2m(0)]/2$. Their fine-grid values are approximately 1.1005634×10^{-5} and 5.20289×10^{-7} ; the residual $R_d = O_d - dm'(0) \simeq -4.31 \times 10^{-10}$ is small but not an exact identity or a certified Taylor remainder. The resolved positive odd change agrees with the local direction. This is a directional response of the fixed nominal actuator, not an identification of rotation as its cause: a free complex accelerator $q_{tt} = \exp[i(\kappa + \delta)t]$ already admits a nonzero detuning slope. More generally, uniform initial-phase averaging selects the normal causal response, whose work derivative is a lag-weighted imaginary response kernel; the carrier and field response enter it jointly. The two-grid guards are diagnostic, and the inherited profile, boundary and linearization limitations persist. No thermal, new-memory-mediator or efficiency conclusion follows.

Prescribed finite phase coherence. A fluctuating drive phase can be specified without identifying it with thermal noise. Let the finite-duration source be $S_\theta(t) = e^{i\theta(t)}S_d(t)$, where the initial phase is uniform and independent and $\theta(t) - \theta(\Delta) = \sqrt{2D} B_{t-\Delta}$ during the pulse, with B a

standard real Wiener process independent of that initial phase. Then $\mathbb{E}e^{i[\theta(t)-\theta(s)]} = e^{-D|t-s|}$ and the anomalous source correlation vanishes. Every realization retains the source's L^2 budget. This does not fix the work delivered to the field.

For the fixed finite real-linear response used above, let $m(\delta)$ denote the phase-averaged source work when the *entire* prescribed source is multiplied by $e^{i\delta(t-\Delta)}$. Its phase-diffusion mean obeys

$$m_D = \int_{-\infty}^{\infty} \frac{D}{\pi(D^2 + \delta^2)} m(\delta) d\delta, \quad D > 0.$$

This follows from the characteristic function of the Cauchy distribution and the quadratic response kernel. At finite time the finite-dimensional propagator is bounded; compact source support and finite L^2 norm give absolute integrability and justify interchanging the integrals. The same weighting convolves the finite-pulse spectral power and preserves its integrated weight. Since the weight is even, the odd detuning response cancels. The positive local slope found above therefore does not determine whether phase diffusion increases or decreases mean work. Nor does a single symmetric detuning pair settle this question.

For computation of this mean work, the lag factor $e^{-D(t-s)}$ can instead be included in the auxiliary generator $A - DI$, evaluated for two genuine source quadratures. With state coordinates (q, p) , both equations acquire $-D$ terms; the work output remains $p + i\omega q$, although now $\dot{q} = p - Dq$. These auxiliary states are not stochastic field trajectories, and their endpoint energies are not ensemble energies. This reduction concerns the specified work observable and does not insert physical friction into the original field equations.

A Cauchy-distributed constant detuning and a Wiener phase process share the relevant two-point statistics but not their trajectory laws. Their quadratic response means agree; the auxiliary damped endpoint is a different object. Ideal Wiener phase also has no finite ultraviolet cutoff. Thus this identity is a finite-model interpretation, not a continuum certificate, a temperature assignment, or a derivation of an environmental bath. Physical bandwidth and coupling require separate specification before experimental use.

One finite-coherence comparison. We fixed $D = 1$, corresponding to one pulse duration for the exponential phase-correlation time, and compared it with $D = 0$. The two genuine quadratures of each auxiliary response were evolved on the same two discretizations as the detuning test. The finer grid gives mean source works $m_0 = 1.133373236 \times 10^{-4}$ and $m_1 = 2.173314375 \times 10^{-4}$. Their difference, $1.039941139 \times 10^{-4}$, is positive on both grids and exceeds the predeclared diagnostic guard 6.63×10^{-9} . The undiffused adapter reproduces the earlier work values, and the modified auxiliary balances include $-2D$ times each quadratic form. Free and restoring-oscillator controls test both the work output and the placement of damping in both auxiliary coordinates.

This is an increase in mean work supplied by the prescribed external source at fixed source L^2 budget. It is not a gain in energetic efficiency or a demonstration of self-sustained oscillations. The guard combines grid differences and implementation residuals; it is not a certified error bound, and space and time resolution change together. The nominal profile, finite boundary and linear-response limitations persist. Only one nonzero diffusion constant is compared: no noise optimum, temperature, nonlinear formation or lifetime conclusion is established. The same source-output reduction with output p gives an additional observable. In the undamped physical finite model, the rotating quadratic generator satisfies $\dot{H} = 2 \operatorname{Re}\langle p, S_\theta \rangle$. For zero initial perturbation, its ensemble endpoint mean therefore equals the half-sum of the two auxiliary integrals of this *original* source rate. It does not equal the auxiliary endpoint H . Extracting these already stored integrals gives, on the finer grid, $\mathbb{E}H(T) = 2.09020168 \times 10^{-4}$ for $D = 0$ and $2.93069596 \times 10^{-4}$ for $D = 1$. Both grids show this approximately 40% increase; this is a descriptive follow-up without a new certified comparison bound or field integration.

This rotating quadratic generator need not be positive, and its global value does not measure energy localized near the Q-ball. Full laboratory endpoint energy additionally requires the physical ensemble charge, including the anomalous state-exchange contribution. The latter is not given by

the exchange evaluated on the damped auxiliary states. Thus neither the source-work increase nor this rotating-generator mean establishes laboratory energy retention or improved confinement.

Correlated instantaneous frequency. A separate prescribed drive model replaces Wiener phase increments by $\theta(t) = \theta_0 + \int_{\Delta}^t \xi(s) ds$, with an independent uniform initial angle and stationary zero-mean Ornstein–Uhlenbeck frequency $\mathbb{E}\xi(t)\xi(s) = (D/\tau_c)e^{-|t-s|/\tau_c}$. Its phase correlation at lag $u \geq 0$ is

$$C(u) = \exp\{-D[u - \tau_c(1 - e^{-u/\tau_c})]\}.$$

This retains the pathwise source norm and fixes the long-time phase diffusion constant D ; the instantaneous frequency variance is D/τ_c . The Wiener kernel is recovered as $\tau_c \rightarrow 0$ at fixed D . Finite τ_c removes its linear short-lag cusp, but does not impose a hard bandwidth cutoff or define a thermal bath. Pointwise larger phase correlation need not give larger work against an oscillatory response.

Small correlation-time limit. For a fixed finite spatial operator and fixed $D \geq 0$, let $\tau = \tau_c$ and write the exact mean work as $m_\tau = \int_0^T W(u)C_\tau(u)du$, with real lag kernel W and $m_D = \int_0^T W(u)e^{-Du}du$. If $|W(u)| \leq M$, then

$$|m_\tau - e^{D\tau}m_D| \leq \frac{e^{D\tau}DM\tau^2}{1 + D\tau}, \quad m'_+(0) = Dm_D.$$

If W is continuous at zero, the present source-work observable has $W(0) = 2B_{\text{src}}$, where $B_{\text{src}} = \int_0^T \|S(t)\|^2 dt$ is the exact nominal source budget. Consequently, $m_\tau = m_D + Dm_D\tau + (D^2m_D/2 - 2DB_{\text{src}})\tau^2 + o(\tau^2)$. These local asymptotics neither determine the comparison at $\tau = 0.1$ nor imply global monotonicity, a thermal interpretation, or a continuum error bound.

For the fixed choice $D = 1$, $\tau_c = 1/10$, put $a = 1/10$. The mean source work has the signed representation $m_{\text{OU}} = e^a \sum_{n \geq 0} (-a)^n m_{1+10n}/n!$, where m_λ is the two-quadrature work obtained from the auxiliary generator $A - \lambda I$. This is not a positive mixture of dissipative states. The selected truncation retains $n = 0, \dots, 8$. If $B \geq 2 \int_{t>s} |K_{\text{work}}(t, s)| dt ds$, its analytical work-tail bound is

$$\epsilon_{\text{tail}} \leq B \frac{10}{9} \frac{(1/10)^9}{9!} \frac{1}{1 - 1/100}.$$

The finite-model bound uses a positive reference energy norm, a coefficient-envelope check and an exact-rational source-norm bound for frozen nominal binary coefficients. This bounds the omitted series, not floating exponential evaluation, signed-sum roundoff or RK integration. Those errors and the combined space/time sensitivity require separate assessment; auxiliary endpoint energies are not ensemble energies. The first finite-correlation implementation passed its scalar controls but failed the preselected modified quadratic-balance tolerance on the coarser field grid. That implementation therefore yielded no accepted finite-correlation field result, and its finer grid was not run. The analytical series-tail estimate did not repair this numerical acceptance failure, and the original outcome and thresholds remain unchanged. A separate calculation with a different integrator and a prospectively specified temporal-sensitivity comparison is reported below.

A separate finite-correlation pulse comparison. A separately specified two-stage Gauss calculation retained the prescribed source, its normalization, $D = 1$, $\tau_c = 0.1$ and the signed nine-term expansion. It split integration steps at the two known source-window joins and used both genuine quadratures for every auxiliary rate, plus a zero-source control. Three fixed integrations compared $(h, \Delta t) = (0.04, 0.005)$, $(0.04, 0.0025)$ and $(0.02, 0.0025)$ through $T = 3$ at $R = 20$. All 108 recorded acceptance checks passed; this comparison was made against the previously stored Wiener-source work on the corresponding spatial grid, not against an auxiliary endpoint energy or a newly repeated Wiener run.

On the finer spatial grid, the truncated OU mean source work was $2.1932622828 \times 10^{-4}$, compared with $2.1733143747 \times 10^{-4}$ for the stored Wiener result. The difference, $1.9947908110 \times 10^{-6}$ (approximately 0.918%), exceeded the predeclared diagnostic guard $1.2658579115 \times 10^{-8}$ and had the same positive sign on the coarser grid. The guard includes the analytical series-tail allowance and measured temporal, spatial, adapter and accounting contributions; it is not a certified total-error bound. The fixed coarse-grid time comparison does not directly measure fine-grid temporal error. This is a bounded increase in mean source work for one prescribed drive in the frozen finite-domain model, not evidence of retained energy, heating, formation, improved efficiency or experimental realizability.

Mean injected work — fixed nominal model

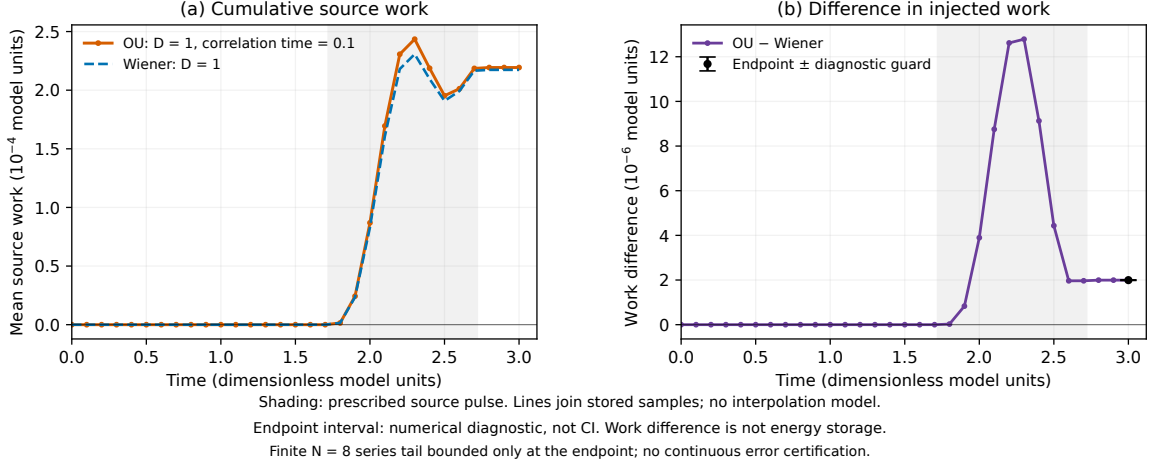


Figure 10: Cumulative mean source work for prescribed OU and Wiener phase drives on the $h = 0.02$ grid (left), and their difference (right). The OU curve is the signed nine-term mean of the two genuine quadrature works per auxiliary rate; the Wiener curve uses the previously stored calculation. Both use integrated original source work, not auxiliary endpoint or ensemble energy. Straight lines join 31 stored times, with no additional field integration for the figure. The endpoint marker spans the predeclared diagnostic guard; it is neither a confidence interval nor a certified total-error bar. All quantities are in dimensionless model units.

One exterior-domain sensitivity check. A separate calculation at fixed $h = 0.04$, $\Delta t = 0.0025$ and $T = 3$ extended $R = 20$ to $R = 30$, retaining byte-identical source values and interior coefficients and comparing with the stored $R = 20$ calculation. Across the 31 shared sample times, the absolute-weighted majorant for the change in the same-method OU-minus- $n = 0$ source-work contrast was 1.14×10^{-19} , below the predefined diagnostic threshold 1.27×10^{-9} . This is agreement at floating-point roundoff scale, not absolute accuracy at 10^{-19} ; common spatial, temporal and profile errors can remain. The check does not establish an infinite-domain limit, long-time behavior or control between samples, and neither replaces the earlier Wiener baseline nor reduces the original comparison guard.

Signed contributions from source-time separations. A separate response-kernel calculation uses the fixed $h = 0.04$, $R = 20$ spatial operator and both genuine quadratures of each source profile. After averaging the uniform initial phase, its mean source work is $m[C] = \int_0^1 W(u)C(u) du$, where u is the separation of two source times, not the observation time. The kernel can have either sign, with $W(0) = 2B_{\text{src}}$. On the finest of three lag meshes (400, 800, 1600 subdivisions), with independent overlap quadratures of orders 16 and 32, the four prescribed quarter-interval

contributions to $\int_0^1 W(u)[C_{OU}(u) - e^{-u}] du$ are, in units of 10^{-6} ,

$$(+23.13074873, -9.08245885, -11.01641674, -1.03681136).$$

Their signed sum is $1.99506177879 \times 10^{-6}$, with summed heuristic refinement envelopes 6.98×10^{-14} ; these are not a certified error bound or confidence interval. The shortest-lag contribution exceeds the three negative later-lag contributions (Fig. 11). Closure to the stored same-grid Gauss OU and Wiener-rate works passes the predeclared diagnostic allowances. The separate shift of the Gauss Wiener-rate work relative to the old RK Wiener baseline is -8.47524×10^{-10} ; it is not assigned to any lag bin. This calculation is not the $h = 0.02$ comparison above and does not exactly decompose its mixed-integrator result. A lag bin integrates pairs of source times, not instantaneous positive or negative power. All separations are at most one model time unit, below the nominal period $2\pi/\rho \simeq 3.44$; the response involves the full generator, not an isolated BIC or a long ringdown. No stored-energy, formation or stabilization conclusion follows.

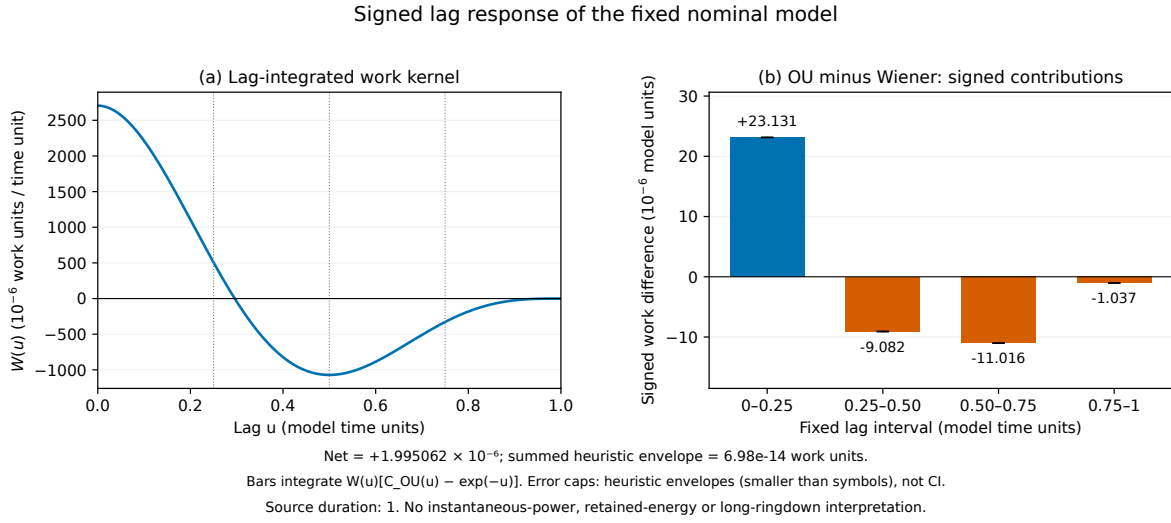


Figure 11: Stored finest-grid signed lag kernel $W(u)$ (left) and four fixed integrals of $W(u)[C_{OU}(u) - e^{-u}]$ (right), for the frozen $h = 0.04$ model. Lines connect the stored lag nodes without smoothing. The caps and reported net envelope are heuristic numerical sensitivity measures, too small to resolve at this scale, not confidence intervals or certified errors; no continuous kernel error band is asserted. The positive net source-work difference retains all three negative contributions. These are source-time-pair contributions, not a time history of instantaneous power or retained energy.

Scalar radiation pressure has been studied in a related one-dimensional Q-ball model [3]; its rates are not imported into our three-dimensional problem.

Several Q-balls of the same nonlinear field can influence one another, but a sum of isolated stationary profiles is not automatically a stationary multi-centre solution. Separation, relative phase and incident frequencies must be specified. We have not derived a self-consistent environmental noise spectrum, an interaction with every particle species, or a mechanism that maintains these excitations indefinitely.

If an additional dynamical mediator is introduced, its spatially nonlocal static response must be distinguished from temporal memory. Matching a homogeneous effective potential does not generally match the nonlocal static interaction. Such extensions are not included in Eq. (1).

An additional open channel also changes the localization condition. For a real-frequency mode with independent propagating asymptotic channels, every propagating tail must vanish separately. A cancellation of the original outgoing amplitude therefore does not establish an embedded bound state of the extended system. Nor does a zero of a leading-order source overlap establish this: the coupled background and mode can change at higher order. These are conditions for a future coupled calculation, not results for an extension of the model studied here.

B Exploratory concentration and constrained transport

A distinct formation hypothesis is that an initially disordered scalar field amplifies a band of spatial wavelengths and subsequently fragments. Condensate fragmentation and a scale selected by growing perturbations are established questions in other Q-ball models [6]. They do not establish formation in our initial-data family or select the BIC frequencies. A future comparison should vary narrow-band and broadband initial spectra at matched total energy and charge, measure growth and outgoing flux, and distinguish transient concentrations from persistent localized charged objects. A connected high-density region in a periodic box should also be checked for closed paths with nonzero winding around the box. A region with such a path is not an isolated density island at that threshold. Absence of winding alone would still not establish vacuum isolation, stationarity or stability; threshold and resolution dependence must be checked. Total charge conservation forbids production of net charge from neutral data; opposite charges or a compensating background would have to be accounted for. No universal integer multiple of a fundamental wavelength is derived here.

Even persistence alone can be biased by the preparation. For the potential in Eq. (1), put $S = |\phi|^2$, $\Pi = \partial_t \phi$, $Q = 2 \operatorname{Im} \int \phi^* \Pi d^3x$ and $q = |Q| > 0$. For finite total energy $E = \int (|\Pi|^2 + |\nabla \phi|^2 + \mathcal{U}(S)) d^3x$, the elementary identity

$$E - q = \int \left[|\Pi - i \operatorname{sgn}(Q) \phi|^2 + |\nabla \phi|^2 - S^2 + \frac{1}{2} S^3 \right] d^3x$$

implies $\int S^2 d^3x \geq q - E$ when $E < q$. Since $\mathcal{U}(S) \geq S/2$, it also gives $\operatorname{ess\,sup} S \geq (q - E)/(2E) > 0$. Thus conserved whole-system energy and charge already preclude uniform dilution to arbitrarily small amplitude for such initial data. This elementary bound is not a formation result: it specifies neither a fixed spatial core nor a single stationary Q-ball. Charge transport, fragmentation and continued oscillation remain possible. A formation test must therefore resolve localization, profile and phase dynamics rather than count survival alone. The bound concerns the closed system, not an unbalanced subvolume or a dissipative boundary model.

A separate finite-time radial relaxation study. A subsequent project study of the same potential evolved four prescribed spherically symmetric Gaussian initial states to $T = 1000$, with an absorbing exterior. Its predeclared diagnostic compared the magnitude of the charge inside $r < 60$ with the stationary-family charge at the frequency estimated from the central phase, allowing a relative difference below 10%. This tests charge–frequency consistency, not equality of the evolving field to a stationary solution. The matched-width state passed at $T = 500$ and the broader state at $T = 1000$. The narrower state failed and retained about 2% of its initial charge inside the measurement region. A larger state remained strongly pulsating and also failed this diagnostic. Consequently, the combined formation criterion fixed before the runs was not met. The successful initial states already had whole-system $E/|Q| < 1$; they do not demonstrate capture of initially unbound radiation. Two jointly refined spatial and temporal resolutions gave the same classifications, but absorber reflection was not measured. These reported finite-time radial observations neither establish exact relaxation nor test nonspherical stability; a charge-retention plateau alone is not a binding proof. The original plan and result are identified in the reproducibility inventory. They have received a document-scope review, not an independent reproduction of the trajectories.

One angular direction during chirped concentration. To examine a freedom excluded by radial evolution, we evolved one prescribed quadrupolar tangent on a time-dependent inward chirp. This is an exploratory formation diagnostic on a different background from the stationary BIC studied above. In the units of Eq. (1), the radial initial data are $F(r) = (3/5)^{3/2} \exp(-r^2/50) \exp(i\beta r^2)$, $\phi_b(r, 0) = F(r)$ and $\partial_t \phi_b(r, 0) = 0.8iF(r)$, with $\beta = 0.015$. Their whole-space analytic values are

$Q \simeq 240.55177$ and $E/Q \simeq 1.038669965 > 1$; hence the preceding $E < |Q|$ non-dilution argument does not force a remnant. The calculation uses a radial Dirichlet box $R = 80$ and ends at $T = 32$.

For $\delta\phi = \eta(r, t)Y_{20}$, with real unit-normalized Y_{20} , put $w = r\eta$ and $S_b = |\phi_b|^2$. The laboratory-frame tangent equation is

$$w_{tt} = w_{rr} - \frac{6w}{r^2} - (1 - 4S_b + \frac{9}{2}S_b^2)w - (-2 + 3S_b)\phi_b^2\bar{w}. \quad (57)$$

We fix $w(r, 0) = r^3(-1/25 + 2i\beta)F(r)$ and $w_t(r, 0) = 0.8iw(r, 0)$, the regular $O(r^3)$ quadrupolar deformation. The nonlinear radial background is advanced simultaneously, without angular backreaction. Its stored diagnostics reproduce the earlier chirp evolution; this replay is a control, not independent formation evidence.

Centered second differences and RK4 are compared at the fixed choices

$$(h, \Delta t) = (0.1, 0.02), (0.05, 0.02), (0.1, 0.01).$$

Origin polynomial, independent free-evolution and conjugate-coupling controls precede the calculation. Our positive reduced diagnostic is

$$N_h^2 = h \sum_{j=1}^{J-1} \left[|w_{t,j}|^2 + \left(1 + \frac{6}{r_j^2}\right) |w_j|^2 \right] + \frac{1}{h} \sum_{j=0}^{J-1} |w_{j+1} - w_j|^2, \quad r_J = 80. \quad (58)$$

It is not a conserved perturbation energy. Positive core, exterior and collar contributions use fixed intervals $[0, 6]$, $[6, 76]$ and $[76, 80]$, with shared-node half weights and complete edge contributions.

On the finer spatial grid, the largest *sampled* $N_h(t)/N_h(0)$ is 1.356829 at $t = 21.6$; its value at $T = 32$ is 1.139233. Samples are separated by 0.8, so this is not a continuous-time supremum. The deformation concentrates in the core and subsequently redistributes outward (Fig. 12). Common-grid endpoint differences, measured in Eq. (58) and divided by the base-grid initial norm, are 0.0062849 for spatial refinement and 4.0021×10^{-7} for timestep halving. These are sensitivity measures, not certified error bounds. The sampled collar contribution satisfies $N_{\text{collar}}^2(t)/N_h^2(0) \leq 10^{-6}$, the prescribed contamination threshold. This single direction shows transient amplification, not a stability classification: the background remains nonstationary, and other radial shapes, angular sectors, finite amplitudes and later times remain untested.

Leading angular feedback on core-charge transport. A separate second-variation calculation follows $\phi_\varepsilon = B + \varepsilon\eta Y_{20} + \varepsilon^2\Xi + O(\varepsilon^3)$ and evolves the forced spherical mean $\zeta = \langle \Xi \rangle_\Omega$. Here Ξ is half the second derivative, and ζ is an angular mean, not the coefficient of normalized Y_{00} . Only this mean is needed alongside the first tangent for the angle-integrated charge through second order. The initial family is the volume-preserving ellipsoidal deformation $r_\varepsilon^2 = e^{-2c\varepsilon}(x^2 + y^2) + e^{4c\varepsilon}z^2$, $c = \sqrt{5/(16\pi)}$, with an explicitly differentiated finite-grid normalization fixing total charge. For Q_2 , the coefficient of ε^2 , define

$$b = \frac{Q_{2,\text{core}}(32) - Q_{2,\text{core}}(0)}{Q_0}$$

in the same fixed core $r < 6$, box $R = 80$ and time window. The three $(h, \Delta t) = (0.1, 0.02), (0.05, 0.02), (0.1, 0.01)$ calculations give $b = 0.08737135, 0.08736583, 0.08737135$, respectively; the spatially refined value exceeds the predeclared heuristic guard 2.864×10^{-5} . This guard combines measured spatial, temporal and charge-balance sensitivities with a fixed floor; it is not a certified error bound or confidence interval. Stage-integrated physical charge flux checks the core-charge difference, rather than defining it from the endpoint.

The interpretation requires the initial redistribution (Fig. 13). On the finer spatial grid, $Q_{2,\text{core}}(0)/Q_0 = -0.10587389$, whereas $Q_{2,\text{core}}(32)/Q_0 = -0.01850805$. The radial reference itself has positive net core inflow over this window: its core fraction increases from 0.58949606

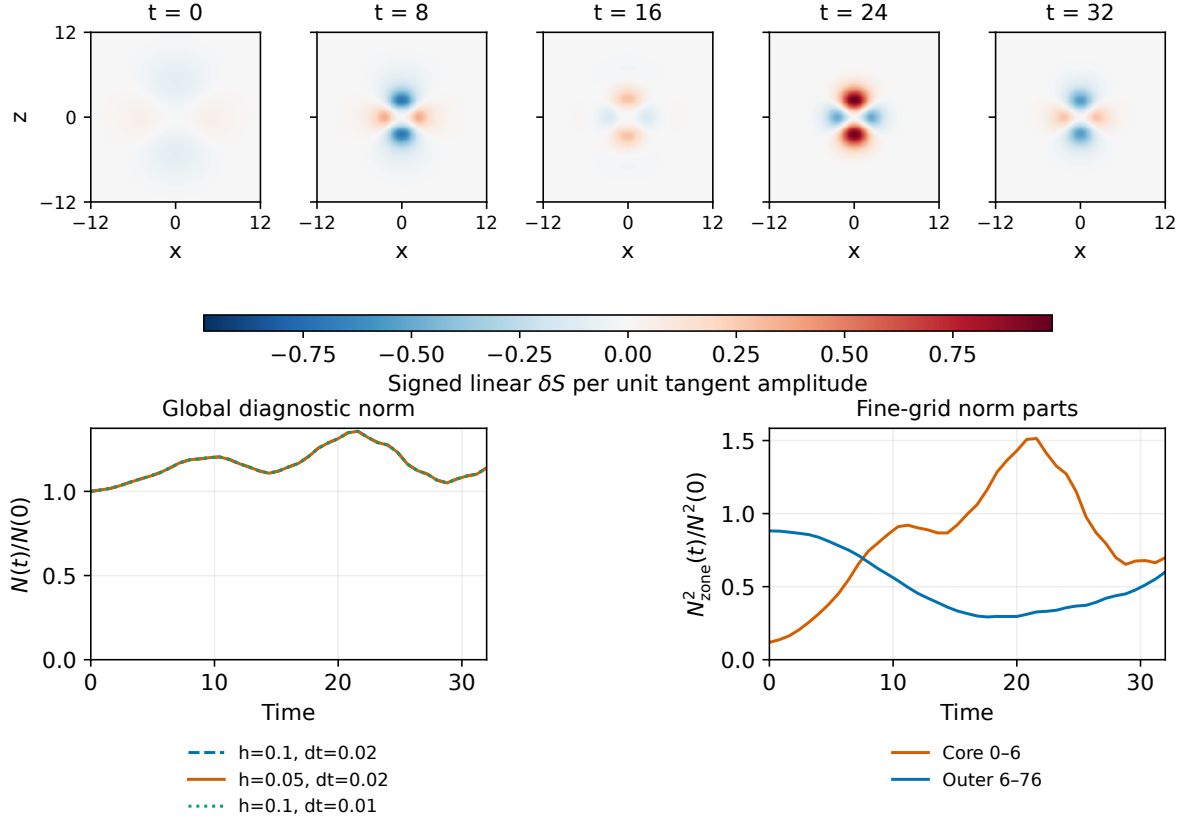


Figure 12: Angular response of the inward chirp. Top: the actual signed linear density variation $\delta S = 2 \operatorname{Re}(\overline{\phi_b} \eta) Y_{20}$ from fine-grid snapshots at $t = 0, 8, 16, 24, 32$, in the fixed xz window $[-12, 12]^2$. Radial nodal values are interpolated for rendering; the common symmetric colour scale is per unit tangent amplitude. Bottom: global norm for all three discretizations, and fine-grid core/exterior squared-norm contributions divided by the *global initial* $N_h^2(0)$. These are linear tangent data, not a finite-amplitude nonlinear three-dimensional evolution or a physical energy partition.

to 0.65267983. The positive second-order net-transport coefficient recovers much of an initial core-charge deficit but leaves the final correction negative; it does not show more final charge retained than in the radial reference. The sampled response is nonmonotonic, with a transient positive core correction before the negative final value; the decision uses the fixed endpoint $T = 32$, not a selected intermediate maximum. The preparation fixes charge, not energy: $E(\varepsilon, 0) = E_0 + \varepsilon^2 E_2 + O(\varepsilon^3)$ with $E_2/E_0 = 0.0224455$ on that grid. These are second-order Taylor coefficients at zero deformation, not a finite-amplitude nonlinear three-dimensional evolution, a binding comparison at equal energy, or evidence of stationary Q-ball formation. As a subsequent analytical interpretation, the untruncated continuum family has $E_{\text{grad}}(\varepsilon) = E_{\text{grad},0}[2e^{-2c\varepsilon} + e^{4c\varepsilon}]/3$, while its kinetic and potential integrals remain unchanged. Thus its positive $E_2 = 4c^2 E_{\text{grad},0}$ is a geometric preparation cost; this continuum identity neither replaces the stored discrete E_2 nor constitutes a new acceptance criterion or determines the transport sign.

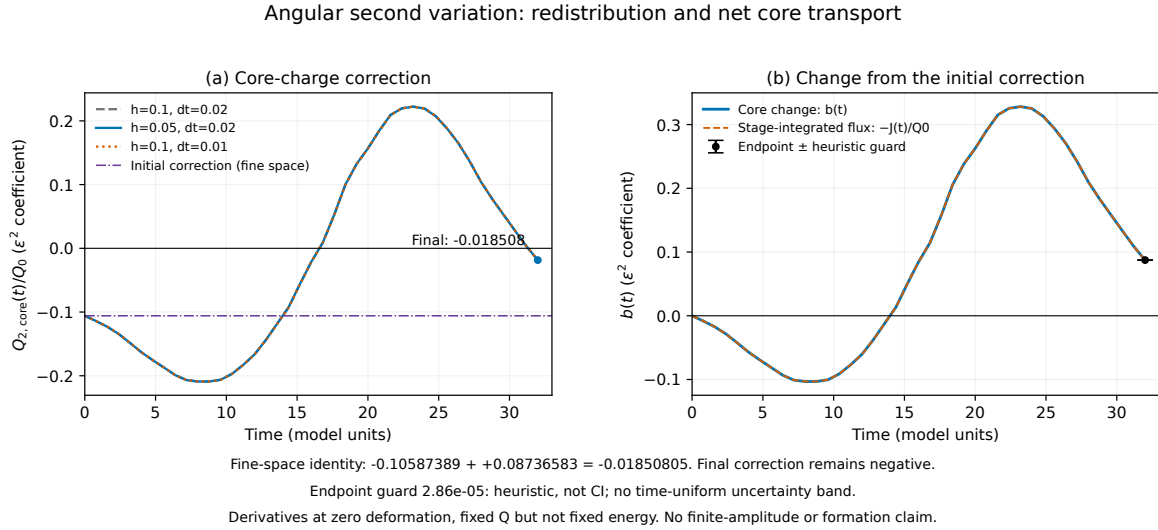


Figure 13: Second-order angular response from stored samples. Left: $Q_{2,\text{core}}(t)/Q_0$ for all three discretizations, with the initial fine-space correction marked separately. Right: the initial-subtracted coefficient $b(t)$ and the negative, independently stage-integrated outward charge flux divided by Q_0 , on the fine-space grid. These routes check the same charge balance, not separate physical replications. Straight lines join 41 stored samples; no finite ε is reconstructed. The final core correction remains negative despite positive net transport. The endpoint guard is heuristic, not a confidence interval, certified error or time-uniform uncertainty band.

Compensating the initial energy through second order. A further, separately specified family keeps the same first angular variation but replaces the initial chirp by $\beta(\varepsilon) = \beta_0 + \beta_2 \varepsilon^2$. On each grid, $\beta_2 = -E_2/(\partial_\beta E_0)$ is determined from the preparation energy, without using the later transport. The normalized family fixes total charge exactly and cancels the initial energy change through order ε^2 ; it does not assert exactly equal energies at finite deformation. Only the homogeneous radial chirp tangent is newly evolved, and its contribution is combined with the stored angular second variation. The earlier fixed-charge result remains unchanged.

On the spatially refined grid, $\beta_2 = -0.0082897517$ and the normalized net-transport derivative with respect to β is -11.0018872 . Their product adds 0.0912029133 to the previous transport coefficient: $b_{QE} = 0.1785687477$, compared with $b_Q = 0.0873658344$. The base and time-refined values are 0.1785384553 and 0.1785384478 ; the predeclared heuristic guard for the spatially refined transport coefficient is 1.525216×10^{-4} , not a certified error interval. The initial core correction is unchanged, while the final core coefficient $c_{QE} = Q_{2,\text{core},QE}(32)/Q_0$ is now $+0.0726948606$, compared with -0.0185080526 for the previous family. This final coefficient is reported separately from

the net-transport decision; no additional endpoint-success gate was introduced. The compensation also reduces the initial inward radial current through its negative chirp correction. Relative to the radial reference, this energy-compensated family changes geometry and radial phase together; its transport response cannot isolate geometry alone. These coefficients concern the fixed endpoint $T = 32$ in the same finite box, not a finite-amplitude deformed solution, stationary formation or binding. Figure 14 separates the core coefficients from their initial-subtracted transport for the two preparations.

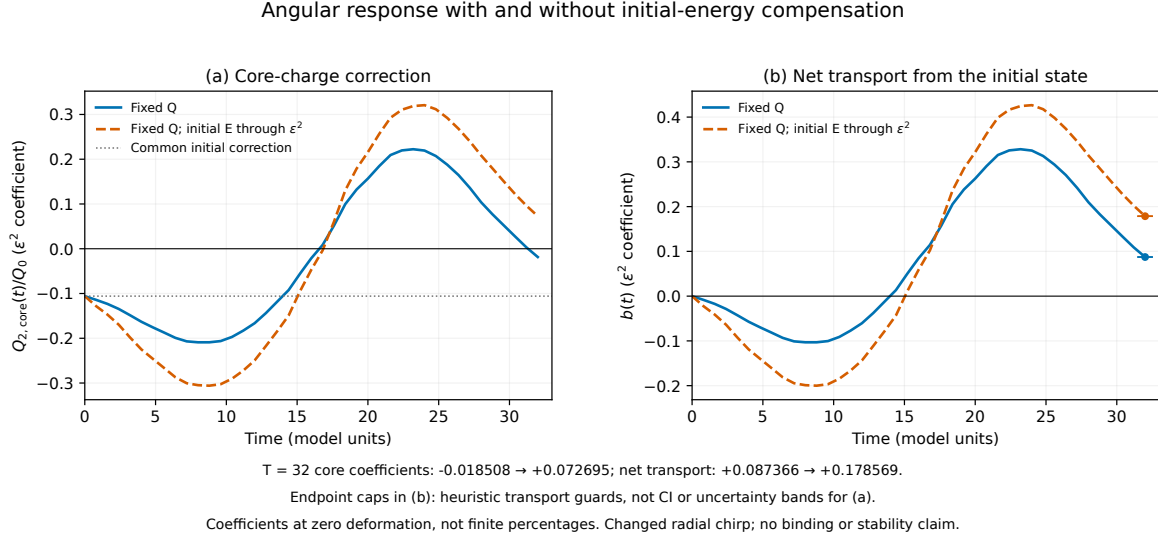


Figure 14: Stored spatially refined second-variation coefficients with and without initial-energy compensation. Left: the core-charge correction; right: its change from the common initial correction. Straight segments join 41 stored samples, without fitting or a new evolution. The original family fixes charge; the compensated family also fixes initial energy through order ε^2 while changing the radial chirp. Neither curve represents a finite deformation or a percentage increase. Endpoint caps on the transport panel are heuristic guards, not confidence intervals, core-coefficient error bars or time-uniform bounds. The sign change of the final core coefficient is not a binding or stability result.

A local two-observable time-shift test. The energy-compensated response need not describe a new stationary state: even an increased endpoint core coefficient could reflect a shifted phase of the transient concentration. A fixed stored-state check at $T = 32$ therefore compares two radial observables, $C = Q_{\text{core}}/Q_0$ and $H = R_{\text{RMS}}^2/6^2$, where R_{RMS}^2 is the second radial moment weighted by $|\phi|^2$ and divided by its spatial integral. Write their second-order coefficients as C_2, H_2 . Here $C_2(32) = c_{QE}$, not the initial-subtracted b_{QE} . A common local shift $t \mapsto t + \varepsilon^2 \tau$ would require $(C_2, H_2) = \tau(\dot{C}_0, \dot{H}_0)$, hence

$$\mathcal{D} = \dot{C}_0 H_2 - \dot{H}_0 C_2 = 0.$$

The moment and its derivative were evaluated directly from the stored Cauchy data, including the norm-denominator variation; no time-series fit or additional field evolution was used. On the fine spatial grid, $H_2 = -1.0936132$ and $\mathcal{D} = 0.0021220476$, exceeding the predeclared heuristic sensitivity guard 6.820019×10^{-5} in inverse model-time units. Thus this pair is inconsistent with one common local clock shift under that diagnostic rule. The charge-only shift coefficient is $\tau_C = C_2/\dot{C}_0 = -9.1439597$, after its separate denominator check: it multiplies ε^2 and is not a measured finite time shift. This is only a two-observable endpoint comparison, conditional on the existing variational outputs; it establishes neither persistence of the core enhancement nor binding, stationary formation or a full-field distance. The sensitivity guard is not a certified error bound.

A finite ellipsoidal response at matched initial charge and energy. We additionally evolved finite, axisymmetric ellipsoidal preparations at $\varepsilon = \pm 0.02, \pm 0.04$, with initial charge and full discrete energy matched to each discretization's radial reference within the declared root tolerance. Unlike the preceding second variation, this tests finite initial data rather than energy compensation only through order ε^2 . At the fixed endpoint $T = 32$ define

$$D(e) = \frac{C(+e) + C(-e) - 2C(0)}{2e^2}, \quad C = Q_{\text{core}}/Q_0.$$

This is a symmetric final core response, not initial-subtracted net transport. On the spatially refined grid, $D(0.02) = 0.0727746617$ and $D(0.04) = 0.0730141978$, against the new same-grid, same-timestep variational reference $c_{\text{ref}} = 0.0726948531$. The prescribed heuristic envelope at $e = 0.02$ is 1.2055721×10^{-4} ; the discrepancy 7.9808621×10^{-5} plus this allowance meets the predeclared one-percent agreement criterion. This envelope is not a certified error bound or a confidence interval. The resolved-remainder criterion is not met, so no measured Taylor order is claimed.

The separate core-fraction differences at $\varepsilon = +0.02, -0.02$ are 2.8253326×10^{-5} and 2.9966404×10^{-5} , respectively, both positive in the stored endpoint data; the symmetric envelope is not a separate certified bound for each sign. These are dimensionless charge-fraction differences, not energy percentages; a positive symmetric coefficient alone would not establish both signs. Figure 15 separates them. The preparation changes geometry and radial chirp together. This finite-time comparison establishes neither persistent enhancement, stationary formation, binding nor general three-dimensional stability. The even- ℓ , $m = 0$ evolution excludes non-axisymmetric dynamics.

The complete twenty-trajectory matrix and separate variational reference were assembled from two bounded production assignments: fourteen completed trajectories and the reference were inherited unchanged, and six remaining trajectories were completed separately. The first assignment retains its INCOMPLETE status; the full matrix, not a favourable subset, is used in the comparison.

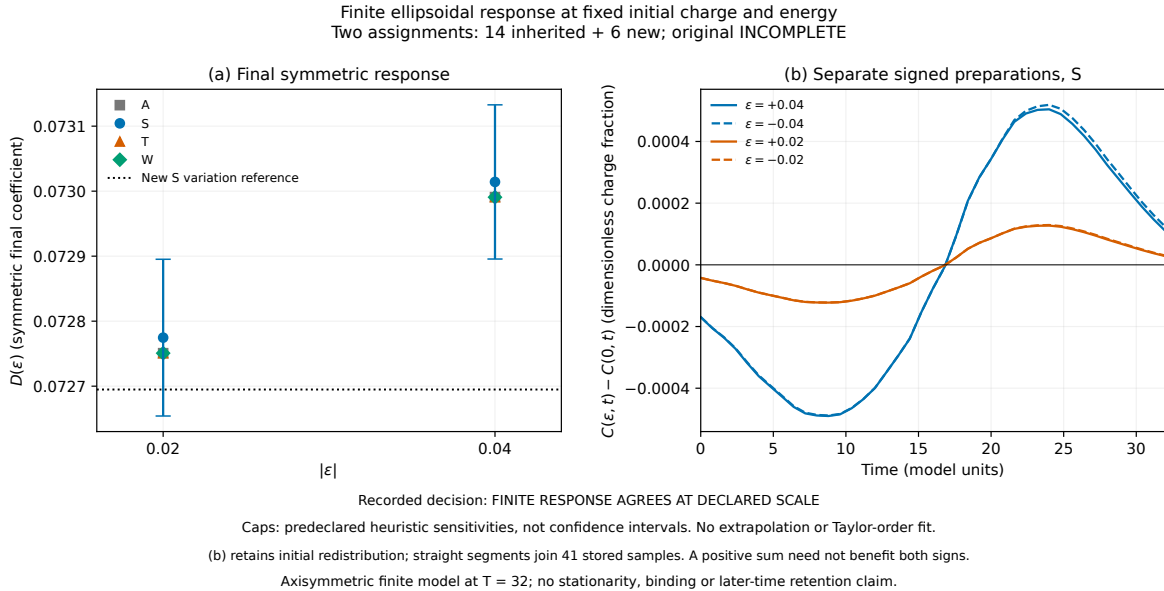


Figure 15: Finite symmetric endpoint coefficients for four discretizations (left) and separate signed core-fraction differences on the spatially refined grid (right). Caps are heuristic sensitivities, not confidence intervals. Straight segments join stored time samples; no extrapolation or Taylor-order fit is used. Initial redistribution is retained rather than subtracted.

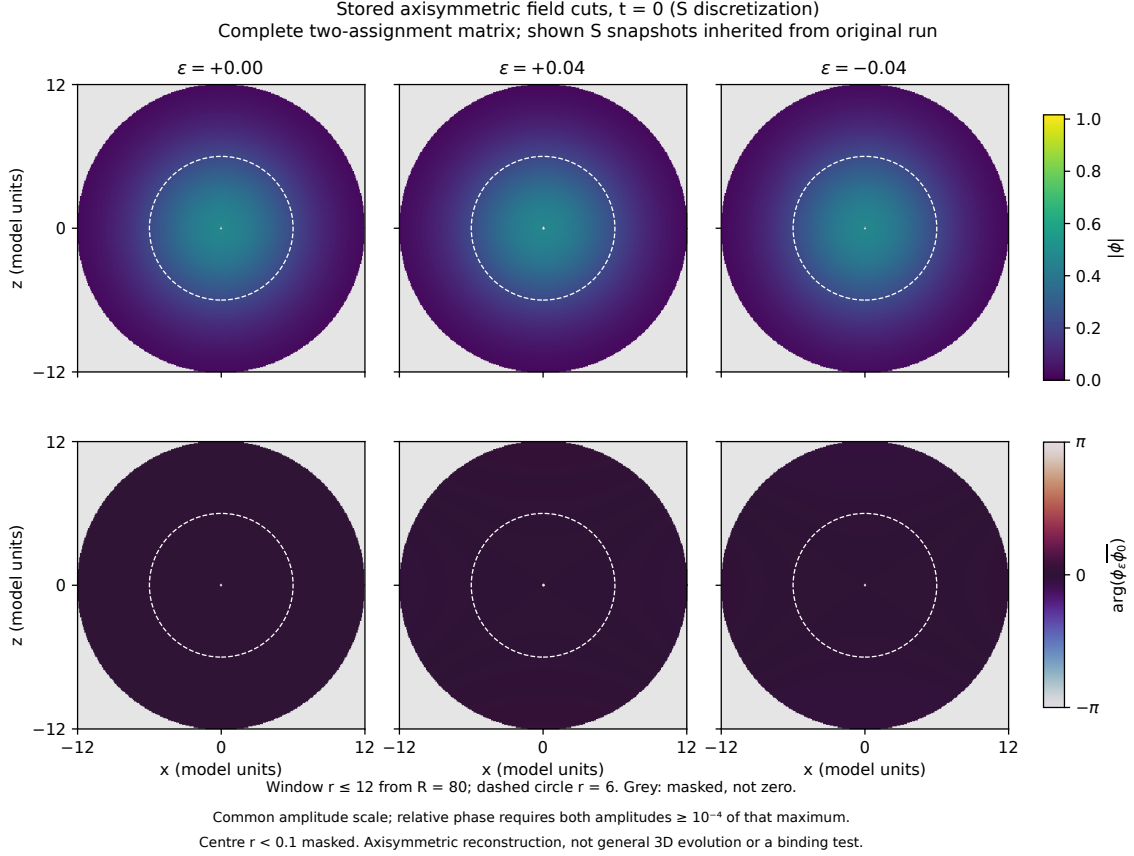


Figure 16: Stored initial amplitude (upper row) and relative phase (lower row) for the radial and $\epsilon = \pm 0.04$ preparations. The meridional cut uses $r \leq 12$ inside the computational domain $R = 80$; the circle marks the measurement radius $r = 6$. The common amplitude scale is shared with Fig. 17. Relative phase is $\arg(\phi_\epsilon \bar{\phi}_0)$ in the common initial phase convention, masked if either amplitude is below 10^{-4} of the shared maximum. The centre $r < 0.1$ is masked; grey is not zero amplitude or phase.

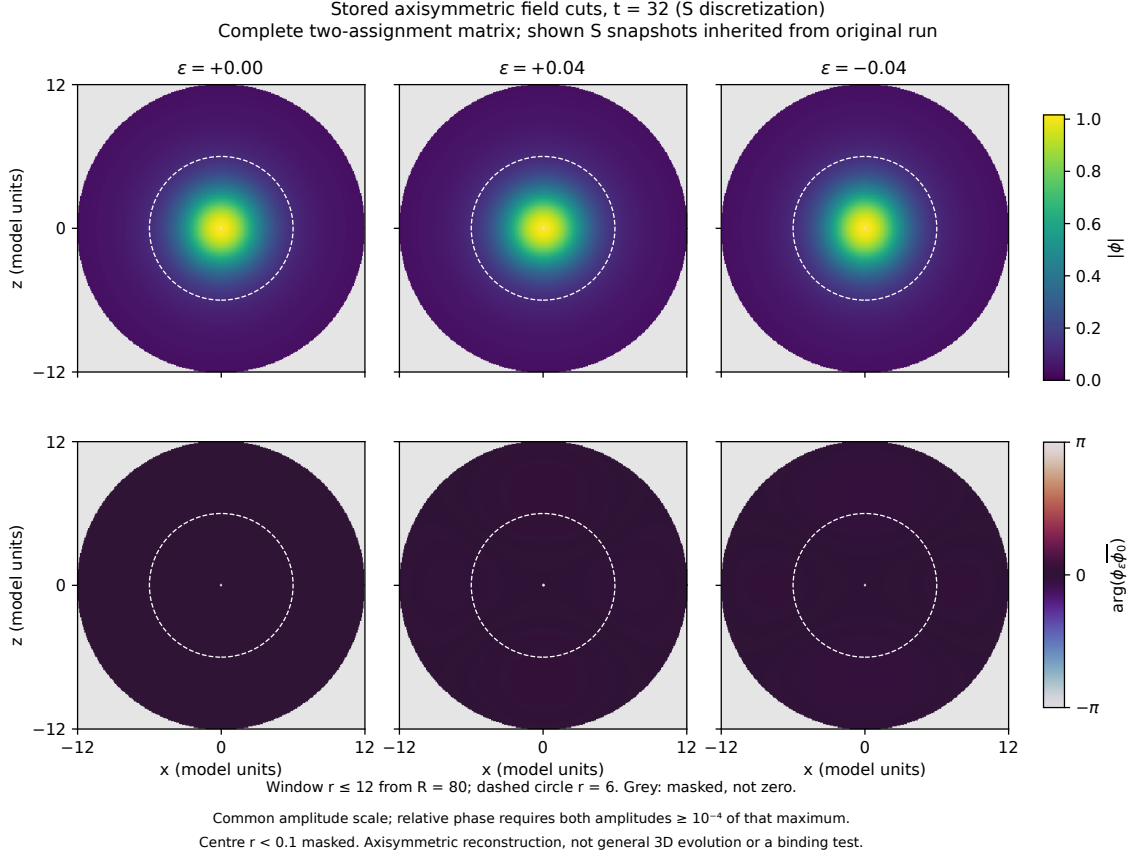


Figure 17: The same stored-field reconstruction at $T = 32$, with the common scales and masks of Fig. 16. Complex radial mode coefficients are interpolated before evaluating the angular basis and phase. These are axisymmetric amplitude/relative-phase cuts, not a general 3D simulation, an absolute-phase observable, or a binding diagnostic. Visually similar colours on the full $[-\pi, \pi]$ scale do not establish exact phase equality.

Phase rotation and the fixed-charge kinetic gap. A similar phase colour does not test rigid harmonic motion. We evaluated necessary conditions on the six stored states with $\varepsilon = 0, \pm 0.04$ at $t = 0, 32$, using the complete $R = 80$ box. For the reduced mode arrays $U = r\phi$, $V = \partial_t U$, write $\|X\|_h^2 = 4\pi h \sum_{j=1}^{J-1} \sum_{\ell} |X_{j\ell}|^2$, $I = \|U\|_h^2$, $K = \|V\|_h^2$ and, for $I > 0$, $\omega_{\text{fit}} = Q/(2I)$. With $A_h(U)$ the actual discrete nonlinear acceleration, define

$$r_v = \frac{\|V - i\omega_{\text{fit}}U\|_h^2}{K + \omega_{\text{fit}}^2 I}, \quad r_a = \frac{\|A_h(U) + \omega_{\text{fit}}^2 U\|_h^2}{\|A_h(U)\|_h^2 + \omega_{\text{fit}}^4 I}.$$

Rigid single-frequency motion requires both numerators to vanish. Initially r_v vanishes by construction, whereas $r_a \simeq 0.0830$; the prescribed phase velocity does not solve the stationary-profile equation. At $T = 32$ the three states give $r_v = 0.0282474$ – 0.0282631 and $r_a = 0.0586355$ – 0.0588005 . These are squared-residual quotients, not amplitude percentages or kinetic-energy fractions. This completed endpoint postprocessing involved no further time integration. It provides neither a core-only classification, a relaxation law, a certified continuum residual bound nor a general 3D stability result.

The unnormalized velocity residual has a separate exact algebraic meaning. In the positive canonical kinetic norm, minimizing velocities at fixed instantaneous U and total Q gives $V_Q = iQU/(2I)$ and

$$E_h(U, V) - E_{Q,h}[U] = J_v = \|V - i\omega_{\text{fit}}U\|_h^2 = K - \frac{Q^2}{4I}, \quad E_{Q,h}[U] = W_h(U) + \frac{Q^2}{4I},$$

where W_h contains the radial and angular gradient plus potential energy. This is a comparison at fixed profile, not a physical projection or measured relaxation: it generally changes regional charge and need not preserve other conserved quantities. It does not minimize the spatial profile, show how energy is radiated, or establish binding. In particular, r_v normalizes J_v by $K + Q^2/(4I)$, not by kinetic or total energy.

Direction of the exterior field. For the same chirp, we analysed the stored radial Cauchy data $u = r\phi_b$, $v = \partial_t u$ at $t = 16, 24, 32$ on all three discretizations. A fixed window χ rises from zero to one on $12 < r < 16$ as $10s^3 - 15s^4 + 6s^5$, $s = (r - 12)/4$; $(\chi u, \chi v)$ is zero-padded onto a periodic analysis interval of length $L = 320$. With Fourier series normalized by $1/M$ for M grid points, both temporal branches are retained: $a_\sigma = (\widehat{\chi u} + \widehat{\chi v}/(\sigma i\Omega))/2$, $\Omega = \sqrt{1 + k^2}$, $\sigma = \pm 1$. The convention $\exp(ikr + \sigma i\Omega t)$ gives group velocity $-\sigma k/\Omega$ and branch energy $8\pi L\Omega^2 |a_\sigma|^2$. The denominator E_{win} below is the *total free spectral window energy*, including the directionless $k = 0$ and unresolved Nyquist bins; it is neither the original nonlinear energy nor its escaped fraction.

The fine spatial grid gives the values below. Define $\epsilon_{\text{nl}} = \|\chi(-2S_b + \frac{3}{2}S_b^2)u\|_2/\|\chi u\|_2$ and $\epsilon_\chi = \|-2\chi' u_r - \chi'' u\|_2/\|\chi u\|_2$, using the radial discrete L^2 norm. These are operator-source quotients in the mass-1 units of the model (dimensionally mass squared), not relative energy errors.

t	$E_{\text{out}}/E_{\text{win}}$	ϵ_{nl}	ϵ_χ
16	0.904702	4.55352×10^{-5}	0.593850
24	0.888886	1.14544×10^{-4}	0.828330
32	0.939306	1.58566×10^{-3}	0.573484

All nine decompositions have $E_{\text{out}} > E_{\text{in}} + E_{\text{Nyquist}}$ and $E_{\text{out}} > E_{\text{win}}/2$. The largest change in the outgoing fraction under spatial refinement is 4.59×10^{-4} , and under timestep halving 1.78×10^{-8} ; these are sensitivity bounds over the sampled cases, not certified errors. The exterior nonlinear source is weak in these units, but the filter source is not: its quotient ranges from 0.57 to 0.83. Zero extension also produces the separately monitored boundary term $u_r(80^-)\delta(r - 80)$, rather than another volume norm. Thus the fixed-window data are predominantly outward-oriented; without control of the accumulated source terms, this does not establish asymptotically free radiation, stationary Q-ball formation or binding of a remaining core.

A prescribed weak-potential extension. In an exploratory, prescribed instantaneous Klein–Gordon–Poisson closure, matching candidates associated with the first two ladder labels persisted at two weak couplings each and shifted to lower ω^2 . These are four tested points, not a uniform robustness region. The observation remains conditional on unresolved Poisson-residual and matching-map regularity checks; it does not establish robustness in the Einstein–Klein–Gordon system or nonlinear stability. Including the potential perturbation here means response within the same instantaneous closure, not full general relativity. The quantity $2|\Phi(0)|$ denotes twice the central potential depth and is not identified with the compactness $2GM/R$.

C Reproducibility and provenance

The spatial and phase figures use the stored T2 reconstruction `dipole-forced/reconstruction/mode-1.npz`, the C0 entry of `RUNDE-09/krein1/profile/PROFILE.json`, and the archived `resonance-20260930/winding/RESULT.json`. The full paths, source hashes, interpolation rules, display thresholds, rendering code and checks are recorded in `coordination/resonance-20260930/paper-v11-figures/`. No new field evolution was performed for these figures. The exploratory two-dimensional comparison is recorded in `coordination/runden-v3/RUNDE-12/leiter2d-praez/`; the additional-channel probe is in `coordination/runden-v3/RUNDE-12/qk1/`. Both are single-house numerical results with subsequent scope reviews, not new interval existence certificates. The numerical-lattice comparison is documented in `RUNDE-23/stille-auf-gitter/` and `RUNDE-24/stille-gitter-2/`, relative to the same rounds directory; the bounded report, symbol and data review is in `coordination/resonance-20260930/grid-robustness-review-20261003/`. This review does not constitute an independent rerun of the grid spectra. The prescribed dipole pulse and its detuning bound are documented in `coordination/resonance-20260930/environment-source-overlap/`, including the result, non-author review and detuning derivation. The conditional monopole comparison is recorded in `dipole-forced/forced/interior-error/error-budget-components/`, relative to the nonlinear-followup directory below. Its result and non-author review bind the full input hashes and directed arithmetic. `BEWEISKETTE-KOMPONENTEN.txt` in the parent directory distinguishes inherited existence and span arguments from the new numerical bounds and the additional continuation hypotheses. This is a project-internal review of the assembled argument, not an independent interval reimplementaion of its entire dependency chain.

The $\beta = 1$ check is documented under `coordination/resonance-20260930/`, in the dossier `planar-wall-beta1-20261003/`. It includes the fixed local search plan, four grid/domain stages, stored complex transmission samples, and non-author code and result reviews. It is a separate modified-potential calculation, not an additional existence certificate for the main $\beta = 1/2$ model.

The accompanying source manifest records the versions read for this manuscript. Proof packages retain their own original code hashes, parameter boxes, interval exports and failed repetitions. The manuscript assembly neither reruns nor relabels those calculations. The first proof package covers radial location C0. The second covers radial location C1 and dipole location C2. Its original dipole repetition failed a sufficient enclosure test; the subsequently expanded parameter box is a separately recorded run, not a retroactively successful preregistered repetition. The primary successful dipole certificate is logically sufficient on its own.

Material	Project-relative source location
Radial certificate	<code>coordination/runden-v3/RUNDE-07/beweis/</code>
Additional certificates	<code>coordination/runden-v3/RUNDE-10/beweis2/</code>
Cross-house source re-views	<code>coordination/resonance-20260930/beweis1-fremdlesung/;</code> <code>coordination/resonance-20260930/beweis2-fremdlesung/</code>
Sequence and predic-tions	<code>coordination/runden-v3/RUNDE-09.md;</code> <code>coordination/runden-v3/RUNDE-09/VORHERSAGEN-MOD2.md</code>
R13 numerical refine-ment	<code>coordination/runden-v3/RUNDE-13/leiter3d-praez/</code> <code>ERGEBNIS.md;</code> <code>coordination/resonance-20260930/paper-r13-review/REVIEW.txt</code>
Spatiotemporal pulse comparison	<code>coordination/resonance-20260930/</code> <code>source-shaping-20261001/;</code> result and independent result review
Factorial actuator com-parison	<code>coordination/resonance-20260930/</code> <code>source-shaping-20261001/factorial/;</code> <code>RESULT.json</code> and <code>RESULT-REVIEW.txt</code>

Material	Project-relative source location
Constant pulse-phase comparison	coordination/resonance-20260930/ source-shaping-20261001/phase-quadrature/; result/review and analysis/figure/SUMMARY.json
Stored mode-defect attribution	coordination/resonance-20260930/ source-shaping-20261001/mode-defect-audit/; result, independent review and proposed insertion
Quadratic endpoint energy	coordination/resonance-20260930/ source-shaping-20261001/phase-quadrature/ endpoint-energy/; RESULT.json and RESULT-REVIEW.txt
Time-integrated power balance	coordination/resonance-20260930/ source-shaping-20261001/phase-quadrature/ endpoint-energy/power-balance/; RESULT.json and RESULT-REVIEW.txt
Phase-dependent source work	coordination/resonance-20260930/ source-shaping-20261001/phase-quadrature/ endpoint-energy/power-balance/phase-work/; coefficient, interval and visualization receipts with reviews
Prescribed temporal phase drift	coordination/resonance-20260930/ source-shaping-20261001/phase-quadrature/ endpoint-energy/power-balance/phase-work/detuning/; out/RESULT.json, fixed plan and non-author result review
Directional derivative and mirror	The preceding detuning directory, directional/; out/RESULT. json, tangent/mirror plan and non-author result review
Prescribed phase diffusion	coordination/resonance-20260930/ source-shaping-20261001/phase-diffusion/; out/RESULT. json, kernel derivations, fixed plan and non-author reviews
Rotating mean and finite correlation	The preceding directory, rotating-mean/ and finite-correlation/; scalar extraction, analytical remainder and failed numerical-acceptance receipt with reviews
Gauss finite-correlation comparison	Relative to the phase-diffusion directory above: finite-correlation/gauss-physical/; out/RESULT.json and RESULT-REVIEW.txt, for the separately accepted finite- domain source-work comparison
Exterior-domain sensitivity	Relative to the preceding Gauss directory: exterior-sensitivity/; out/RESULT.json and RESULT-REVIEW.txt, for the fixed-time, fixed-resolution domain comparison
Signed source-time contributions	Relative to the same Gauss directory: kernel-mechanism/; out/ RESULT.json and RESULT-REVIEW.txt, for the signed lag-kernel diagnostic
Angular chirp diagnostic	coordination/resonance-20260930/ formation-nonlinear-3d/angular-chirp-l2/; prod/RESULT. json, non-author result review and visualization receipt
Exterior window diagnostic	coordination/resonance-20260930/ formation-nonlinear-3d/angular-chirp-l2/ exterior-analysis/; out/RESULT.json, non-author result review and fixed-window specification
Thin-wall derivation	coordination/runden-v3/RUNDE-09/MODELL-DUENNWAND.md

Material	Project-relative source location
Independent planar-wall scattering	coordination/resonance-20260930/ planar-wall-blind-20261002/; BLIND-ENDE.sha256, out-v2/RESULT.json, ERROR-RESULT.json and non-author review; comparison in coordination/runden-v3/RUNDE-23/ tropfen-leiter/ERGENIS.md
Finite ellipsoidal re- sponse	coordination/resonance-20260930/ formation-next-20261002/fixed-energy/ finite-amplitude/; original and rest production records, composite result, independent result review and rendering receipts
Phase/profile diagnos- tics	The preceding directory, rigidity-execution/out/RESULT. json, PHASE-RIGIDITY-RESULT-REVIEW.txt and paper-prep/ KINETIC-GAP-NOTE.txt; stored endpoint measurements and sep- arate fixed-charge algebra
Radial Gaussian relax- ation	coordination/runden-v3/RUNDE-23/bildung-3d/; KARTE.md, frozen plan PLAN.md.eingefroren-20261002-220532, ERGENIS.md; document-scope review in coordination/ resonance-20260930/paper-v38-literature/ BILDUNG-3D-READING.txt
Evidence audit	coordination/resonance-20260930/novelty-audit/ LEITER-STATUS.txt
Quadratic-source check	coordination/resonance-20260930/ bic-nonlinear-followup/
Dipole input and re- sponse	coordination/resonance-20260930/ bic-nonlinear-followup/dipole-forced/

These are local project identifiers, not publicly accessible archival citations. Before submission, the underlying supporting files must be packaged in a durable public supplement with a license and an archival identifier. Their current local availability must not be described as a published independent replication.

D Separate numerical-method controls

The original finite-correlation field calculation remains a numerical acceptance failure, and no accepted OU mean is inferred from it. A separate constant-source control exposed a finite-step defect in the RK4 quadratic balance: an exact finite-step RK4 identity reproduced the stored endpoint and accumulated-rate discrepancies, within the prescribed floating-point comparison tolerances, for 27 case-step combinations sampled at 31 times each. This identifies the numerical mechanism in those controls, not the complete cause of the earlier spatial-field failure. A subsequent two-stage Gauss control evaluated nine constant-system cases (three damping rates and three complex sources) at three time steps. Its recorded acceptance checks passed, including independent exact-state and directly integrated-rate references at the finest step, the prescribed error non-growth checks, and rejection of an intentionally incorrect rate. The conclusion therefore does not rest solely on Gauss’s algebraic quadratic-balance property. These controls alone do not validate the original physical-pulse field calculation or decide the OU-versus-Wiener comparison; the subsequent prescribed-pulse calculation is reported separately in the main text.

A subsequent spatial adapter used a smooth manufactured solution on the actual finite operator with $R = 20$, $h = 0.04$ and 499 interior nodes, through $T = 1$. It covered nine damping rates $\lambda = 1, 11, \dots, 81$ and a separate zero control at time steps 0.005, 0.0025 and 0.00125. Independent full-state references and direct integral references for five original rates and three modified quadratic rates passed their finest-step accuracy checks and prescribed error non-growth checks, with a

non-author result review supporting the recorded outcome. This control alone validates only the manufactured solution, not the prescribed physical pulse, an OU mean, or a continuum limit.

The finite-step comparison and the Gauss control have separate non-author result reviews. Their fixed plans, results and reviews are recorded under `coordination/resonance-20260930/source-shaping-20261001/phase-diffusion/finite-correlation/` `diagnostic-step/`, in `recurrence-check/` and `gauss-control/`, respectively; the spatial adapter’s separate result and review are in `gauss-control/spatial-manufactured/`. The small-correlation-time derivation and its independent paper review are recorded in `gauss-control/`. These local method checks are not additional physical findings or interval-certified integration errors. The subsequent physical-pulse comparison, with its own fixed plan, result, non-author review and figure receipts, is recorded in `coordination/resonance-20260930/source-shaping-20261001/phase-diffusion/finite-correlation/gauss-physical/`. Its `exterior-sensitivity/` subdirectory records the separate $R = 30$ comparison, fixed decision rule, retained $R = 20$ reference and non-author result review. No historical trajectory was rerun for that comparison. The accepted signed-lag calculation, its fixed plan, independent result review and stored-data figure are recorded in `kernel-mechanism/`, another subdirectory of `gauss-physical/`, with `out/RESULT.json`, `RESULT-REVIEW.txt` and `figure/out/`. The restricted weak-potential discussion follows the project model review and its independent algebraic reading in `coordination/resonance-20260930/qstern-paper-review/`, namely `REVIEW-ROOT.txt` and `REVIEW-MODEL-NONAUTHOR.txt`; the latter is not a new audit of the numerical matching claims. The angular second-variation derivation, fixed plan, code and result reviews are in `coordination/resonance-20260930/formation-next-20261002/`; the accepted result is `execution-r2/out/RESULT.json`, and the stored-data figure is in `figure/out/`. The preceding import-only adapter failure remains recorded separately. The subsequent continuum preparation-energy identity and its algebraic review are explanatory checks, not additional prospective gates or physical replications. The separately specified energy-compensated chirp tangent, its derivation, plan and code are in the `fixed-energy/` subdirectory of `formation-next-20261002/`, with the new stored result in `execution/out/RESULT.json`. Its reference angular second variation retains its original result and provenance; it was not evolved again. The energy-compensated comparison figure uses the stored data and receipt in `fixed-energy/figure/out/`. Its separate local-clock check is in `fixed-energy/time-shift/`, including the derivation, fixed endpoint plan and `execution/out/RESULT.json`; it reuses the existing Cauchy states rather than evolving another trajectory.

Contributions and disclosure

The working draft was assembled with OpenAI agents from computational and analytical material developed in the AG Physics project, including an Anthropic-authored thin-wall section and proof packages. AI-assisted source reviews are project-internal checks, not journal peer review. Authorship is assigned to Finn Malte Hinrichsen, Hamburg, at his explicit direction. No institutional affiliation is asserted. Submission and public release remain separate decisions.

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